BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Factors influencing the readiness to tackle the burden of ischemic heart disease in India: a systematic review protocol.

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-047464
Article Type:	Protocol
Date Submitted by the Author:	30-Nov-2020
Complete List of Authors:	Bhattacharjee, Shuvarthi; University of Brighton, School of Health Sciences Neupane, Dinesh; Johns Hopkins University Bloomberg School of Public Health, Epidemiology Phuong, Cao; Ho Chi Minh City University of Medicine and Pharmacy, Pharmacy Yaghmaei , Nima
Keywords:	Ischaemic heart disease < CARDIOLOGY, Health policy < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, HEALTH SERVICES ADMINISTRATION & MANAGEMENT, PUBLIC HEALTH, Cardiac Epidemiology < CARDIOLOGY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

ADMINISTRATIVE INFORMATION

1. Title of the review	**Factors influencing the readiness to tackle the burden of ischemic heart disease in India: a systematic review protocol.**
2. Registration	This systematic review protocol is registered in the International Prospective Register of Systematic Reviews (PROSPERO) with number CRD 42020219490 following method guidelines from Cochrane handbook of systematic reviews of interventions to conduct of this systematic review, reported in line with the PRISMA statement [1,2]. Link: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=219490
3. Keywords	Humans; India; Myocardial Ischemia; Systematic Review; Health Services; Noncommunicable Diseases;
4. Word Count	3598 words excluding abstract & references.
5. Conflicts of Interest	The Author(s) declare(s) that there is no conflict of interest.
6. Acknowledgements	None.
7.Data Statement	No dataset is available for this systematic review yet. However, preliminary search result can be found in the following link: https://www.crd.york.ac.uk/PROSPEROFILES/219490_STRATEGY_20201106.pdf
8. Review Status	Ongoing. At present undergoing searches across electronic databases.

AUTHORS

CONTACT(3a)	
	Mr. Shuvarthi Bhattacharjee (SB) M.Sc. (Public Health) Marie Sklodowska Curie Actions (MSCA) Early Stage Researcher, School of Health Sciences, University of Brighton (Falmer Campus), Brighton, BN19PH.UK. Telephone No.: +441273644025 E-mail: S.Bhattacharjee@brighton.ac.uk Orchid Identifier: https://orcid.org/0000-0002-6079-5365
	Dr Dinesh Neupane (DN) PhD Research Associate,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	Welch Center for Prevention, Epidemiology and Clinical Research, Department of Epidemiology, Johns Hopkins Bloomberg School of Public Health Baltimore, Maryland, Baltimore, USA. E-mail: dneupan2@jh.edu Orchid Identifier: https://orcid.org/0000-0002-1501-2990
	Ms. Cao Tran Le Phuong (CLP) M.Sc. (Public Health) Researcher, University of Medicine & Pharmacy, 217, Hong Bang Street, Ward 11, District 5, HCMC, Vietnam. Telephone No.: +842838358411 E-mail: caotran_lephuong@yahoo.com
	Mr. Nima Yaghmaei (NY) MIH Independent Consultant, Toronto-M5M3S3, North York, Ontario, Canada. Telephone No.: + 32498068096 E-mail: nimayagh@gmail.com
Corresponding Author	Mr. Shuvarthi Bhattacharjee (SB) Marie Sklodowska Curie Actions (MSCA) Early Stage Researcher, School of Health Sciences, University of Brighton (Falmer Campus), Brighton, BN19PH.UK. Telephone No.: +441273644025 E-mail: S.Bhattacharjee@brighton.ac.uk Orchid Identifier: https://orcid.org/0000-0002-6079-5365
CONTRIBUTIONS (3b) (Authors)	SB is the guarantor of the review. SB & NY conceptualized the study. A search strategy will be created and run on relevant platforms by SB. All search results will be retrieved by SB from appropriate databases, trial registry (for details of primary studies in reviews) & grey literature. CLP & NY will validate the search strategy prior to implementation and check the retrieved search results. SB, NY and CLP will screen articles and any discrepancies for final inclusion of Full- text articles will be resolved by DN. All authors will be involved in data extraction, quality appraisal, data synthesis and formulation of the final draft.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

AMENDMENTS (4)	Important protocol amendments post registration will be recorded, updated in PROSPERO and included in knowledge dissemination.

SUPPORT	
Sources (5a)	No Financial support was received for this study. Authors are academic students or staff in universities & civil society organizations. Access to Healthcare Databases Advanced Search (HDAS) will be provided by NHS Openathens, UK for searching of databases. Access to Web of Science will be provided by the University of Brighton (UK). Data Management support will be sought from COVIDENCE® systematic review software, Veritas Health Innovation, Melbourne, Australia[3].
Sponsor (5b)/Funding	This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.
Role of sponsor/funder (5c)	Not Applicable.

Methodology for this Protocol
We followed the PRISMA-P guidelines which recommend items to address in a systematic review protocol.

Abstract

Introduction: Ischemic Heart Disease (IHD) is one of the leading causes of death & disease burden in India affecting all age groups. To reduce the deaths and tackle the burden of existing IHD the government approach has been mostly through the National Health Policy (2017) and National Program for Prevention & Control of Diabetes, Diabetes, Cardiovascular diseases and Stroke (NPCDCS). This paper offers a protocol for the systematic review of studies exploring the factors influencing service readiness of the public health system of India to tackle the burden of IHD.

Methods and Analysis: : Electronic databases of EMBASE (Ovid), AMED(Ovid), HMIC(Ovid), BNI(ProQuest), CINAHL(EBSCO), EMCARE(Ovid), PsycINFO(ProQuest), Medline/PubMed and Web of Science (Clarivate Analytics) will be searched till 1st October, 2020 for primary studies. Grey literature will be accessed through OpenGrey, TRIP Medical, WHO Database, MoHFW Website, OGD Platform India and Google Scholar (only first 10 pages). Primary studies meeting the eligibility criteria & grey literature published in English between 2000 to 2020 will be included. Data will be analysed through a conceptual framework and primary outcome will constitute both quantitative and qualitative data. The quality of included studies will be assessed based on study design. Data will be managed on the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

COVIDENCE® platform. All authors will be involved in data extraction, quality appraisal, data synthesis and formulation of the final draft.

Ethics and dissemination: This study being a systematic review does not involve any clinical trial, primary data collection or empirical study involving humans or animals. Therefore, no ethical permissions were sought by reviewers.

Registration details: This systematic review protocol is registered in the International Prospective register of Systematic Reviews (PROSPERO) with number CRD 42020219490.

Link: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=219490

Strengths and limitations of this study

- To our knowledge, this is the first systematic review on the factors influencing readiness of the public health system of India to tackle the burden of Ischemic Heart Disease.
- The study can inform policymakers, academicians, NGOs and healthcare administrators with understanding of the existing gaps in the health system.
- This study can provide options to optimise existing health system resources of India to extend better quality of care for the patients.
- Our use of a deductive approach with a conceptual framework for data analysis may exclude some findings which might be a limitation.
- India being a vast country with a decentralised system of health governance might have varied levels of service readiness across states making it difficult to generalise study findings for all contexts.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

INTRODUCTION

6. Rationale

Ischemic Heart Disease (IHD) forms an integral part of the broad Cardiovascular Diseases (CVDs) domain concerned with disease of the heart caused by narrowing of the coronary arteries that supply blood to the heart muscles[4]. In 2017, IHD alone claimed approximately 1.5 million deaths in India with a mortality rate of 111.6 deaths per 100000 populations[5]. Between 2007 and 2017, deaths due to IHD increased significantly by 49.8% in India and in the year 2017 contributed to 7.71% of the total Disability Adjusted Life Years (DALYs) in 2017, making it the leading cause to deaths & disease burden in the country[5]. In fact, evidence shows that when numerically combined for deaths and disability, contribution of IHD was more than 40%, the highest during the period of 2007-2017[5]. IHD affected all age groups including the working population of the country, leading to their premature deaths[5].

High burden of disease & deaths due to IHD can have significant socio-economic implications for India. Specifically, the working population of India who are earning members within households may suffer premature deaths, leading to loss of individual lives and resulting in economic instability for their dependents. Moreover, healthcare expenditure is majorly met through out of pocket (OOP) expenses, presently at 63% of current health expenditure (CHE) of India, pushing households towards financial impoverishment[6]. In such a situation, tackling the disease burden and reducing deaths due to IHD seems necessary to avert loss of individual lives and prevent possible financial impoverishment of their households.

In order to reduce the deaths and tackle the burden of IHD in India the government approach has been mostly through context centric tactics of tackling the Non-Communicable Diseases (NCDs) through nationwide health sector policy & programs[7]. In agreement with the Global Action Plan for Prevention & Control of NCDs 2013-2020, endorsed by the World Health Assembly (WHA) in 2013, India has adopted 25 indicators and 9 voluntary goals to reduce 25% of premature deaths due to NCDs by 2025[8,9]. Premature mortality associated with IHD is indicated as unconditional probability of dying due to IHD between the age bands of 30-70 years[8]. In India, the translation of this global action plan at national level has occurred through the National Health Policy (NHP-2017) and the policy vision is being implemented across states through multiple approaches by the National Program for Prevention & Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS)[(10,11)]. Additionally, apart from this, India's flagship program on universal health coverage (UHC), Ayushman Bharat, has plans to convert 150000 Primary Health Centres (PHCs) into Health & Wellness Centres (HWCs) to strengthen primary care services that can prevent & treat NCDs including IHD[12]. To support the cause, the NHP-2017 planned to raise the General Government Health Expenditure (GGHE) to 2.5% of the GDP from the usual government allocation of 1.2% of the GDP in previous years. Moreover, NHP-2017 has suggested NPCDCS to

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

strengthen secondary & tertiary care services, along with primary care NCD interventions, such as 'Best Buys'[13].

However, while the plans and commitments have been put in place, the challenges of the existing disease burden of IHD, upscaling premature deaths due to IHD, overwhelmed public healthcare system, scarcity of health workers and the poor co-operation between states and the central government in India's decentralized health administrative structure appear to be detrimental towards tackling the disease burden of IHD [14-16]. Therefore, it becomes necessary to explore the factors influencing readiness of the public health system of India in order to tackle IHD based upon the commitments made through the NHP-2017 & other relevant national programs to tackle the increasing number of cases & deaths due to IHD. Readiness here is defined as the cumulative availability of components required to provide services[17].

In a nutshell, this is an attempt to explore whether the existing public health system is capable of delivering the government commitments on national health policy and national NCD program to the people of India, through the existing public health system infrastructure, in order to tackle the mortality & morbidity due to IHD.

Studies addressing similar issues for India are mostly limited to small-scale empirical studies which fail to give national level insights[18,19]. Our preliminary search on this topic in Cochrane Database of Systematic Reviews, PROSPERO, JBI Evidence Synthesis, CRD/DARE Database and Campbell Collaboration revealed no systematic reviews, scoping reviews, rapid reviews or published protocols related to India concerning readiness of health system. However, there were reviews focussed on health system components for LMIC settings, acute care settings concerning LMIC countries, monitoring of cardiac patients and risk factors associated with IHD in Asian Indians (included those of Indian origin living outside India) which bear no similarity to our study[20-24]. Moreover, data from such studies considered LMIC context in a totality which made it difficult to find any specific data for India. This study is more relevant in the present COVID-19 crisis whereby health services, especially tertiary care cardiac units having ventilators, are over capacity and overwhelmed across the country [25].

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

8.Eligibility Criteria

i) Population, or participants and conditions of interest

People born and brought up in India residing presently in India either living with or at the risk of Ischemic Heart Disease. Primary studies related to such population type published in peer reviewed journals and grey literature in English published between 2000 and 2020 will be included.

ii) Interventions or exposures

Relevant primary studies in peer reviewed journals & grey literature on national programs of NCD, National Health Policies (NHP-2012 & NHP-2017), interventions concerned with both prevention & control of Ischemic Heart Disease in India will be included if published in English between 2000 to 2020.

iii) Comparisons or control groups None.

iv) Outcomes of interest

Data of both quantitative & qualitative nature related to health system preparedness or readiness to prevent & tackle the burden of Ischemic Heart Disease in India.

v) Setting

Primary, Secondary & Tertiary Care settings in Public Health System of India.

vi) Study Designs

All forms of study design of both quantitative & qualitative nature will be included.

vii) Time Frame: None.

viii) Reporting Characteristics:

Years Considered: 2000 to 2020; Language: English; Publication Status: Peer Reviewed. This process will be done manually. No filters will be applied in search databases.

9. Search Strategy: Electronic databases of EMBASE (Ovid), AMED(Ovid), HMIC(Ovid), BNI(ProQuest), CINAHL(EBSCO), EMCARE(Ovid), PsycINFO(ProQuest), Medline/PubMed and Web of Science (Clarivate Analytics) will be searched with use of appropriate Boolean operators (AND, OR & NOT) till 1st October, 2020. No filters or limits will be used at this stage. All electronic databases (except Medline/PubMed and Web of Science) will be accessed through the Healthcare Databases for NHS in England (HDAS) of the NHS Openathens, UK. HDAS mirrors search results of platforms like Ovid, EBSCO and ProQuest to retrieve results from these databases.

Trial registers like PROSPERO, Cochrane Database of Systematic Reviews, Campbell Collaboration and CRD/DARE Database will be searched to avoid duplication of existing knowledge on the topic and identify relevant primary studies from available systematic reviews.

Moreover, citation tracking, footnote chasing, bibliography scan and citation alerts (on web of science) will be conducted for a comprehensive search[26]. A combination of free texts and corresponding synonyms of terms comprising but not limited to “Ischemic Heart Disease” AND “Health System” AND “Readiness”

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

AND "India" will be made for the initial search and appropriate preidentified MeSH Terms ('Emtree Terms' for Embase) in PubMed/Medline will be included to make a comprehensive search strategy across the electronic databases (Table 1 added as a representative example). Thesaurus terms appropriate to the databases will be used to account for the variation in such terms across databases.

Grey literature will be accessed through OpenGrey, TRIP Medical, WHO Database, MoHFW Website, OGD Platform India and Google Scholar (only first 10 pages).

Detailed list of information sources & representative search result for this study are given in ***'Supplementary Material A: Information Sources & Representative Search Strategy'.***

STUDY RECORDS

10a. Data management

COVIDENCE® systematic review software, Veritas Health Innovation, Melbourne, Australia (available at : www.covidence.org) will be employed to manage the studies retrieved and initiate the process of systematic review. On this software duplicate articles will be rejected in order to gather a set of unique articles for this study. All article screening activities, selection of final set of articles for inclusion in the study after examination of full texts of eligible studies, data extraction and risk of bias assessment will be carried out on COVIDENCE®. References will be managed and stored on Endnote®. All forms of article selection, data extraction, quality appraisal, risk of bias assessment will be performed and a 'PRISMA flow chart', based on PRISMA guidelines for systematic review, will be created on the COVIDENCE® platform.

10b. Selection process

Initially, all duplicate articles will be checked by three reviewers (SB, NY, CLP) individually through a double blinded process to assure a rigorous review approach and rejected before identifying a unique set of articles for this study on the COVIDENCE® platform.

Afterwards, the screening of title and abstracts of potential articles will be conducted by the same reviewers in a similar double blinded manner to assure rigor of the study[27]. In case of any conflict related to the inclusion of an article, it will be temporarily included in the next step for further evaluation.

In the next step, full-text articles will be examined based on the inclusion and exclusion criteria before final inclusion in the study. Reasons for the inclusion and exclusion of articles will be clearly identified and documented by each reviewer independently on ***Supplementary Material B*** (Form for Screening of full text articles based on inclusion & exclusion criteria with documented justification). Assuring three reviewers will allow for a consensus-based conflict resolution in case there is uncertainty raised about inclusion of an article for the study. Moreover, the DN will also be consulted before finalising the final set of included articles for this study.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

10c. Data Collection Process

Data will be extracted based on ***Supplementary Material C*** (Data Extraction Form for Included Studies) by three reviewers (SB, NY & CLP) independently in a blind manner for each included study. SB will randomly select half of the included studies which will be given to DN to extract data independently. This will ensure rigor of the study and a consensus-based approach for approving the final content of the data extraction form. Any disagreements will be resolved through mediation with DN. The data extraction form will also offer content on the risk of bias and quality assessment of included studies. As the final data synthesis will involve a conceptual framework, all relevant quantitative data will be converted into meaningful qualitative data.

11. Data Items

The descriptive data items that will be collected constitute : (i) Characteristics of the study including country & year of publication, research aim & participant characteristics; (ii) Methodological characteristics including study design, data/information collection methods and data analysis; (iii) Results associated with readiness of the public health system to tackle the burden of ischemic heart disease along with concluding remarks from the article. The data collection items will also include risk of bias and quality assessment of included individual articles based on appropriate tools. For this study we wish to create simplification of quantitative data by transforming them to meaningful qualitative units to accommodate them into the conceptual framework opted for our study. Moreover, only relevant data will be extracted based on the needs of the study.

We will extract data from those studies concerned with primary care facilities where facility assessment can provide data pertaining to facility readiness even if readiness concerning ischemic heart disease is not mentioned. This is because treatment of Ischemic Heart Disease and its risk factors are integrated at the primary care level along with treatment of other communicable and non-communicable diseases[28].

12. Outcomes and prioritization

Outcome of this study will constitute both quantitative and qualitative data retrieved from the included articles to inform health system factors that influence the readiness of India's public health system to tackle the burden of Ischemic Heart Disease.

This will include data at all levels of care that informs health service delivery, health workforce, health information system, access to essential medicines, health financing and health system governance concerning the present topic.

In terms of quantitative data, data from facility readiness assessment, health facility assessment, WHO Service Availability & Readiness Assessment (SARA) data, surveys, baseline study of health facilities and monitoring & evaluation data of health facilities along with data from the national NCD program of India will serve to provide information on the outcome.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

In terms of qualitative studies, data from in-depth interviews, focus group discussions, researcher observations of facilities and data from the national health policies will serve to understand the outcomes better.

13. Risk of bias in individual studies

All eligible studies at the outcome stage are expected to have methodological variation in approach and therefore, we will assess their quality & risk of bias at this stage. In order to minimize the risk of bias and evaluate the quality of included studies the Cochrane collaboration tool for assessing the risk of bias of Randomized Control Trials (RCTs), Newcastle-Ottawa Scale for cohort & case-control studies, NIH Quality Assessment Tool for Cross-Sectional Studies and Critical Appraisal Skills Program (CASP) tool for qualitative studies will be used based on the methodological design of the individual eligible study[29-32].

We will not reject any study based on assessed quality only the information will be documented in the data extraction and synthesis stage. The risk of bias assessment and quality appraisal will be independently performed by each reviewer before converging to a mutually agreed decision. Any discrepancies will be mediated by lead author DN.

National Health Policies of India, national NCD program guidelines and raw primary data from the OGD Platform cannot be assessed for risk of bias and quality appraisal. They will be excluded from this process of evaluation.

14. DATA

14 (a) Data Synthesis

The extracted data will be analysed through the WHO Framework for Action conceptual framework followed by summarization of findings to answer the research questions[33]. We do not expect to conduct a meta-analysis as we do not anticipate many articles in which meta-analysis may be appropriate. Additionally, the inclusion of qualitative studies will likely lead to a heterogeneous outcome which is not suitable for meta-analysis. Moreover, we will be transforming the quantitative data to meaningful qualitative units for assimilation into the conceptual framework for our study. The WHO Framework for Action rather serves as a conceptual lens for this study. Therefore, we will be deductive in our approach assuming all health system readiness issues are inclusive within this conceptual framework. The conceptual framework was chosen through mediation with all reviewers and moreover, straight forward demarcation of health system components allows for clear interpretation[34]. We also considered other conceptual frameworks; however, they were either overly complex for the purposes of this study or considered factors beyond the health system that were not relevant for the present study [35-39].

14 (b) Sub-Group Analysis

At present, considering the research objectives, we do not plan to undertake any sub-group analysis. However, should we find after the data extraction stage that certain sub-groups or sub-sets like gender,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

rural/urban population, health professionals in varied levels of care etc. can provide unique insights on the topic under consideration then we might undertake such sub-groups/ subsets analysis.

14 (c) Meta-Bias(es)

We do not plan to conduct any assessment of meta-bias(es) as the expected studies included for this systematic review might be of varied study design making it difficult to assess such bias such as 'publication bias across studies' or 'selective reporting within studies'[40,41].

14 (d) Confidence in cumulative evidence

At present we do not have plans to assess the confidence in cumulative evidence as this study does not report effectiveness of clinical outcome(s).

15. Presenting or reporting the results

For this systematic review the process of screening to final selection of articles for inclusion in the study will follow the flow diagram in accordance with the PRISMA-P guideline comprising of retrieval of potential studies from mentioned sources, screening the articles, evaluating the eligibility of the articles based on inclusion/exclusion criteria and further analysis of included studies[42]. Data collection will be through a data extraction form with outcomes to extract data related to health system readiness. Data will be summarized in tables with details of study characteristics, methodological characteristics and individual study results. We will also provide data of included and excluded studies with documented justification of their inclusion or exclusion.

16. Patient and public involvement

The present study is a systematic review not an empirical study and therefore, there is no direct involvement of patient or public in this study.

17. Ethics & Dissemination

17 (a) Ethical Issues

This study being a systematic review does not involve any clinical trial, primary data collection or empirical study involving humans or animals. Therefore, no ethical permissions were sought by reviewers.

17 (b) Publication Plan

We plan to publish the preliminary results of this review in conferences and proceed on to publish the final article in peer reviewed journal once completed. We hope to publish both the protocol & systematic review in open access journals to benefit academic communities, policymakers and other public health stakeholders with the ambition to improve the readiness of the health to tackle the burden of Ischemic Heart Disease in India and elsewhere.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

References:

1. Higgins JPT, Green S (editors). Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0 [updated March 2011]. The Cochrane Collaboration, 2011. Available from www.handbook.cochrane.org.
2. Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.
3. Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia. Available at www.covidence.org .
4. Institute of Medicine (US) Committee on Social Security Cardiovascular Disability Criteria. Cardiovascular Disability: Updating the Social Security Listings. Washington (DC): National Academies Press (US); 2010. 7, Ischemic Heart Disease. Available from: https://www.ncbi.nlm.nih.gov/books/NBK209964/.
5. Institute for Health Metrics and Evaluation (IHME). Findings from the Global Burden of Disease Study 2017. Seattle, WA: IHME, 2018. Available from: https://vizhub.healthdata.org/gbd-compare/
6. World Health Organization (WHO). Global health expenditure database. Last updated August 20, 2020. http://www.who.int/health-accounts/ghed/en/ (accessed Sept. 18, 2020).
7. Ministry of Health & Family Welfare (MoHFW, India). National Health Portal (2019, November 5). Non-communicable Diseases. Retrieved from https://www.nhp.gov.in/healthlyliving/ncd2019 (accessed Sept 18, 2020).
8. World Health Organization. (2013). Global action plan for the prevention and control of noncommunicable diseases 2013-2020.
9. Ministry of Health & Family Welfare (MoHFW, India). National Action Plan and Monitoring Framework for prevention & control of Noncommunicable Diseases (NCDs). Retrieved from https://www.iccp-portal.org/system/files/plans/India%20-%20National_Action_Plan_and_Monitoring_Framework_Prevention_NCD_2013.pdf (accessed on Sept 18, 2020).
10. Ministry of Health & Family Welfare (MoHFW, India). National Health Policy; 2017. Available from: http://www.cdsco.nic.in/writereaddata/National-Health-Policy.pdf (accessed on Sept. 18, 2020).
11. Ministry of Health & Family Welfare (MoHFW, India). Directorate General of Health Services (DGHS). National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke. Retrieved September 18, 2020, from https://dghs.gov.in/content/1363_3_NationalProgrammePreventionControl.aspx

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12. Gopichandran, V. Ayushman Bharat National Health Protection Scheme: An Ethical Analysis. ABR 11, 69–80 (2019). https://doi.org/10.1007/s41649-019-00083-5
13. World Health Organization. (2017). Tackling NCDs: 'best buys' and other recommended interventions for the prevention and control of noncommunicable diseases (No. WHO/NMH/NVI/17.9). World Health Organization.
14. Narain J. P. (2016). Public Health Challenges in India: Seizing the Opportunities. Indian journal of community medicine: official publication of Indian Association of Preventive & Social Medicine, 41(2), 85–88. https://doi.org/10.4103/0970-0218.177507
15. Tiwari, R., Negandhi, H., & Zodpey, S. P. (2018). Health Management Workforce for India in 2030. Frontiers in public health, 6, 227. https://doi.org/10.3389/fpubh.2018.00227
16. Lakshminarayanan S. (2011). Role of government in public health: Current scenario in India and future scope. Journal of family & community medicine, 18(1), 26–30. https://doi.org/10.4103/1319-1683.78635
17. Abazinab, S., Woldie, M., & Alaro, T. (2016). Readiness of Health Centres and Primary Hospitals for the Implementation of Proposed Health Insurance Schemes in Southwest Ethiopia. Ethiopian journal of health sciences, 26(5), 449–456. https://doi.org/10.4314/ejhs.v26i5.6
18. Jayanna K, Mony P, B M R, et al. Assessment of facility readiness and provider preparedness for dealing with postpartum haemorrhage and pre-eclampsia/eclampsia in public and private health facilities of northern Karnataka, India: a cross-sectional study. BMC Pregnancy Childbirth. 2014; 14:304. Published 2014 Sep 4. doi:10.1186/1471-2393-14-304
19. Nirula SR, Naik M, Gupta SR. NHS vs Modicare: The Indian Healthcare v2.0. Are we ready to build the healthier India that we envisage? J Family Med Prim Care. 2019;8(6):1835-1837. doi: 10.4103/jfmpc.jfmpc_309_19
20. Marin, Tania S.1,2; Walsh, Sandra3; May, Nikki4; Jones, Martin3; Gray, Richard5; Muir-Cochrane, Eimear1; Clark, Robyn A.1 Screening for depression and anxiety among patients with acute coronary syndrome in acute care settings: a scoping review, JBI Evidence Synthesis: September 2020 - Volume 18 - Issue 9 - p 1932-1969 doi: 10.11124/JBISRIR-D-19-00316
21. Likka, Melaku Haile1,2; Handalo, Dejene Melese1,3; Weldsilase, Yetnayet Abebe4; Sinkie, Shimeles Ololo3 The effect of community-based health insurance schemes on utilization of healthcare services in low- and middle-income countries: a systematic review protocol of quantitative evidence, JBI Database of Systematic Reviews and Implementation Reports: March 2018 - Volume 16 - Issue 3 - p 653-661 doi: 10.11124/JBISRIR-2017-003381
22. Teferra, Meseret N.1; Ramos, Joyce S.2; Kourbelis, Constance2; Newman, Peter2; Fleury, Amanda1; Hobbs, David1; Reynolds, Karen J.1; Clark, Robyn A.2 Electronic textile-based electrocardiogram monitoring in cardiac patients: a scoping review, JBI Database of Systematic Reviews and

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Implementation Reports: October 2019 - Volume 17 - Issue 10 - p 1958-1998 doi: 10.11124/JBISRIR-2017-003989

23. Ninan, Ajitha MSN, RN, APN-C1; Acosta, Juvy MSN, RN, APN-C1; Kulesza, Theodora MSN, RN, APN-C1; Mattis, Patrick DNP, MSCS, RN, BC, CPHIMS1; Holly, Cheryl EdD, RN1 A systematic review of the effectiveness of preventive measures for coronary artery disease in Asian Indians, JBI Library of Systematic Reviews: Volume 9 - Issue 64 - p 1-17 doi: 10.11124/jbisrir-2011-314.
24. Ninan, Ajitha MSN, RN, APN-C[1]; Acosta, Juvy MSN, RN, APN-C[1]; Kulesza, Theodora MSN, RN, APN-C[1]; Mattis, Patrick DNP, MSCS, RN, BC,CPHIMS[1]; Holly, Cheryl EdD, RN[1] A systematic review on the risk factors of coronary artery disease found in Asian Indians, JBI Library of Systematic Reviews: Volume 9 - Issue 48 - p 1-14 doi: 10.11124/jbisrir-2011-455.
25. Kumar, P., & Kumar, M. (2020). Management of potential ventilator shortage in India in view of on-going COVID-19 pandemic. Indian journal of anaesthesia, 64(Suppl 2), S151–S152. https://doi.org/10.4103/ija.IJA_342_20
26. Wright, K., Golder, S., & Rodriguez-Lopez, R. (2014). Citation searching: a systematic review case study of multiple risk behaviour interventions. BMC medical research methodology, 14, 73. https://doi.org/10.1186/1471-2288-14-73
27. Ayala Quintanilla BP, Taft A, McDonald S, et alSocial determinants and maternal exposure to intimate partner violence of obstetric patients with severe maternal morbidity in the intensive care unit: a systematic review protocol. BMJ Open 2016;6:e013270. doi: 10.1136/bmjopen-2016-013270
28. Ministry of Health & Family Welfare, Government of India. National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Disease and Stroke: Operational Guidelines. Strategy. 2010.
29. Higgins Julian P T, Altman Douglas G, Gøtzsche Peter C, Jüni Peter, Moher David, Oxman Andrew D et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials BMJ 2011; 343 :d5928
30. Wells GA, Shea B, O'Connell D, Peterson J, Welch V, Losos M, et al. (2009) The Newcastle-Ottawa Scale (NOS) for assessing the quality if nonrandomized studies in meta-analyses. ; Available from: URL:http://www.ohri.ca/programs/clinical_epidemiology/oxford.htm
31. National Institutes of Health (2014). Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies. Available online at: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools (Accessed November 19, 2020).
32. Critical Appraisal Skills Programme. (CASP) qualitative checklist, https://casp-uk.net/wp-content/uploads/2018/01/CASP-Qualitative-Checklist-2018.pdf (accessed 18 Sept. 2020)
33. World Health Organization. (2007). Everybody's business--strengthening health systems to improve health outcomes: WHO's framework for action.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

34. Mounier-Jack, S., Griffiths, U.K., Closser, S. et al. (2014). Measuring the health systems impact of disease control programmes: a critical reflection on the WHO building blocks framework. BMC Public Health 14, 278. https://doi.org/10.1186/1471-2458-14-278
35. Roberts, M.J., W.C. Hsiao, P. Berman, and M.R Reich. 2003. Getting Health Reform Right. New York: Oxford University Press, page 27
36. De Savigny, D. and Adam, T. (eds.) 2009, Systems thinking for health systems strengthening, World Health Organization, Geneva.
37. Shakarishvili, George, Mary Ann Lansang, Vinod Mitta, Olga Bornemisza, Matthew Blakley, Nicole Kley, Craig Burgess, and Rifat Atun. (2011). "Health systems strengthening: a common classification and framework for investment analysis." Health policy and planning 26, no. 4 : 316-326.
38. Josefien van Olmen, Bart Criel, Wim Van Damme, Bruno Marchal, Sara Van Belle, Monique Van Dormael, Tom Hoerée, Marjan Pirard and Guy Kegels (2010). Analysing health systems to make them stronger.ITG Press, Antwerpen, Belgium.
39. Murray, C. J., & Frenk, J. (2000). A framework for assessing the performance of health systems. Bulletin of the world Health Organization, 78, 717-731.
40. Dalton, J. E., Bolen, S. D., & Mascha, E. J. (2016). Publication Bias: The Elephant in the Review. Anesthesia and analgesia, 123(4), 812–813. https://doi.org/10.1213/ANE.0000000000001596
41. Reid, E. K., Tejani, A. M., Huan, L. N., Egan, G., O'Sullivan, C., Mayhew, A. D., & Kabir, M. (2015). Managing the incidence of selective reporting bias: a survey of Cochrane review groups. Systematic reviews, 4, 85. https://doi.org/10.1186/s13643-015-0070-y
42. Moher, D., Liberati, A., Tetzlaff, J., & Altman, D. G. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. PLoS Medicine, 6(7), e1000097. https://doi.org/10.1371/journal.pmed.1000097.

For Peer Review Only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Material A: Information Sources & Representative Search Strategy			
(i) Electronic Databases:	**Database**	**Accessed From**	**Planned Dates of Coverage**
	Pubmed	NIH Website	From inception till 01.10.2020
	Embase	NHS HDAS Platform	From inception till 01.10.2020
	EMCARE	NHS HDAS Platform	From inception till 01.10.2020
	HMIC	NHS HDAS Platform	From inception till 01.10.2020
	BNI	NHS HDAS Platform	From inception till 01.10.2020
	CINAHL	NHS HDAS Platform	From inception till 01.10.2020
	AMED	NHS HDAS Platform	From inception till 01.10.2020
	PsycINFO	NHS HDAS Platform	From inception till 01.10.2020
	Web of Science	Clarivate Analytics	From inception till 01.10.2020
(ii)Trial Registers	PROSPERO	CRD Database	From inception till 01.10.2020
	Cochrane Database of Systematic Reviews	Cochrane Library	From inception till 01.10.2020
	Campbell Collaboration	Cambell Collaboration Website.	From inception till 01.10.2020

For peer review only

	JBI Evidence Synthesis	Joanna Briggs Institute Website.	From inception till 01.10.2020
	CRD/DARE Database	Centre for Reviews and Dissemination Website.	From inception till 01.10.2020.
(iii) Contact with study authors	Researchgate	Researchgate Website	As per necessity to request for full text articles based on articles published till 01.10.2020.
(iv) Grey Literature Sources	OpenGrey	INIST-CNRS managed Website	01.01.2000 to 01.10.2020
	TRIP Medical	TRIP Database Website	01.01.2000 to 01.10.2020
	WHO Database	World Health Organization Website	From inception till 01.10.2020.
	MoHFW	Ministry of Health & Family Welfare Website.	From inception till 01.10.2020.
	OGD Platform India	Government of India Website on Open Government Data.	From inception till 01.10.2020.
	Google Scholar	Google Scholar Website (only first 10 pages)	From inception till 01.10.2020.

For peer review only

Table 1		
A representative search for CINAHL database performed on NHS Openathens HDAS Platform on 08th August 2020. No Filters or Limits were applied.		
Sl. No.	**Search Terms**	**Results**
1	(India*).ti,ab OR (South Asia*).ti,ab OR (Asia*).ti,ab OR (LMIC).ti,ab OR (LIC).ti,ab OR (Low Income Countr*).ti,ab OR (Low Middle Income Countr*).ti,ab OR (SAARC).ti,ab OR (ASEAN).ti,ab	85642
2	(NCD*).ti,ab OR (non-communicable diseas*).ti,ab OR (chronic disease* OR myocardial ischemia).ti,ab OR (cardio?vascular diseas*).ti,ab OR (ischemic heart diseas* OR heart disease*).ti,ab OR (vascular disease*).ti,ab OR (Diabetes).ti,ab OR (Hypertension).ti,ab OR ("High Blood Pressure").ti,ab OR (Cancer).ti,ab OR (COPD).ti,ab OR (Chronic Obstructive Pulmonary Disease).ti,ab OR (Coronary Artery Diseas*).ti,ab OR (Coronary Heart Diseas*).ti,ab	789662
3	(health system* OR program*).ti,ab OR (health?care system*).ti,ab OR (health planning* OR health service utiliz*).ti,ab OR (health?care planning*).ti,ab OR (health facilit*).ti,ab OR (National Health Policy).ti,ab OR (National Health Program*).ti,ab OR (NPCDCS).ti,ab	565703
4	(prepare* OR health system prepared*).ti,ab OR (ready).ti,ab OR (readiness).ti,ab OR (facility? assess*).ti,ab OR (readiness).ti,ab	77767
5	exp INDIA/ OR exp "ASIA, WESTERN"/	128168
6	exp ASIANS/	32187
7	exp "DEVELOPING COUNTRIES"/	18662
8	exp "LOW- AND MIDDLE INCOME COUNTRIES"/	1361
9	(5 OR 6 OR 7 OR 8)	176910
10	(1 OR 9)	225727

For peer review only

11	exp "CARDIOVASCULAR DISEASES"/ OR exp "HEART DISEASES"/ OR exp "VASCULAR DISEASES"/	584557
12	exp "CORONARY DISEASE"/	54503
13	exp "MYOCARDIAL ISCHEMIA"/ OR exp "ACUTE CORONARY SYNDROME"/	101965
14	exp "CHRONIC DISEASE"/ OR exp "NONCOMMUNICABLE DISEASES"/	64096
15	exp "DIABETES MELLITUS"/	164718
16	exp HYPERTENSION/	78476
17	exp "LUNG DISEASES, OBSTRUCTIVE"/	60675
18	exp NEOPLASMS/	544583
19	(11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18)	1311975
20	(2 OR 19)	1554463
21	(10 AND 20)	50731
22	exp "HEALTH AND WELFARE PLANNING"/ OR exp "HEALTH FACILITY PLANNING"/	171562
23	exp "HEALTH CARE DELIVERY"/	338701
24	exp "NATIONAL HEALTH PROGRAMS"/ OR exp "HEALTH RESOURCE UTILIZATION"/ OR exp "HEALTH SERVICES ACCESSIBILITY"/ OR exp "MANAGED CARE PROGRAMS"/ OR exp "PRIMARY HEALTH CARE"/ OR exp "SECONDARY HEALTH CARE"/ OR exp "TERTIARY HEALTH CARE"/	237596
25	exp "PUBLIC HEALTH ADMINISTRATION"/ OR exp "PROGRAM DEVELOPMENT"/	88044
26	exp "PUBLIC POLICY"/ OR exp "HEALTH POLICY"/	106774
27	exp "HEALTH POLICY STUDIES"/ OR exp "POLICY STUDIES"/	827
28	(22 OR 23 OR 24 OR 25 OR 26 OR 27)	526561
29	(3 OR 28)	959473
30	(21 AND 29)	10200
31	(4 AND 30)	258

For peer review only

For peer review only

Supplementary Material B: Form for Screening of full text articles based on inclusion & exclusion criteria with documented

Basic Inclusion Criteria								SB		CLP		NY		
Sl. No.	Title	Is this peer reviewed?	Is this paper in English?	Is this a primary study?	Published between 2000 to 2020?	Does it concern India?	Article related to the topic?	Reasons for Inclusion	Reasons for Exclusion	Reasons for Inclusion	Reasons for Exclusion	Reasons for Inclusion	Reasons for Exclusion	Final Decision (Yes/No)
1														
2														
3														
4														
5														
6														
7														
8														
9														
10														

For peer review only

Supplementary Material C: Data Extraction Form for Included Studies									
Sl. No.	Title	Author	Year & Country	Research Aims	Study Design	Participants	Methods	Results	Conclusions
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									

For peer review only

Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	1
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	1
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1-2
Contribution	#3b	Describe contributions of protocol authors and identify the	2

For peer review only

		guarantor of the review	
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	2
Support			
Sources	#5a	Indicate sources of financial or other support for the review	3
Sponsor	#5b	Provide name for the review funder and / or sponsor	3
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	3
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	8-9
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	10-12
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	12
Study records -	#11b	State the process that will be used for selecting studies (such	13

For peer review only

selection process		as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	13-14
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	14
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	14
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	15
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	16
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	16
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	16
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	16
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	16
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	17

For peer review only

The PRISMA-P checklist is distributed under the terms of the Creative Commons Attribution License CC-BY 4.0. This checklist was completed on 29. November 2020 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

For peer review only

BMJ Open

Factors influencing the readiness to tackle the burden of ischemic heart disease in India: a systematic review protocol.

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-047464.R1
Article Type:	Protocol
Date Submitted by the Author:	19-May-2021
Complete List of Authors:	Bhattacharjee, Shuvarthi; University of Brighton, School of Health Sciences Yaghmaei , Nima ; HumERA Consulting, Phuong, Cao; Ho Chi Minh City University of Medicine and Pharmacy, Department of Public Health Neupane, Dinesh; Johns Hopkins University Bloomberg School of Public Health, Epidemiology
Primary Subject Heading:	Public health
Secondary Subject Heading:	Health policy, Health services research, Global health
Keywords:	Ischaemic heart disease < CARDIOLOGY, Health policy < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, HEALTH SERVICES ADMINISTRATION & MANAGEMENT, PUBLIC HEALTH, Cardiac Epidemiology < CARDIOLOGY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

ADMINISTRATIVE INFORMATION

1. Title of the review	**Factors influencing the readiness to tackle the burden of ischemic heart disease in India: a systematic review protocol.**
2. Registration	This systematic review protocol is registered in the International Prospective Register of Systematic Reviews (PROSPERO) with number CRD 42020219490 following method guidelines from Cochrane handbook of systematic reviews of interventions to conduct of this systematic review, reported in line with the PRISMA statement. Link: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=219490
3. Keywords	Humans; India; Myocardial Ischemia; Systematic Review; Health Services; Noncommunicable Diseases;
4. Word Count	3674 words excluding abstract & references.
5. Conflicts of Interest	The Author(s) declare(s) that there is no conflict of interest.
6. Acknowledgements	None.
7.Data Statement	No dataset is available for this systematic review yet. However, preliminary search result can be found in the following link: https://www.crd.york.ac.uk/PROSPEROFILES/219490_STRATEGY_20201106.pdf
8. Review Status	Ongoing. At present undergoing searches across electronic databases.

AUTHORS

CONTACT(3a)	
	Mr. Shuvarthi Bhattacharjee (SB) M.Sc. (Public Health) Early Stage Researcher, School of Health Sciences, University of Brighton (Falmer Campus), Brighton, BN19PH.UK. Telephone No.: +441273644025 E-mail: S.Bhattacharjee@brighton.ac.uk Orchid Identifier: https://orcid.org/0000-0002-6079-5365
	Mr. Nima Yaghmaei (NY) MIH Independent Consultant, HumERA Consulting, 115, Esgore Drive, Toronto, Ontario, Canada M5M3S1

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	Telephone No.: + 32498068096 E-mail: nimayagh@gmail.com Orchid Identifier: https://orcid.org/0000-0001-7740-8037
	Ms. Cao Tran Le Phuong (CLP) M.Sc. (Public Health) Researcher, Public Health Department, University of Medicine & Pharmacy, 217, Hong Bang Street, Ward 11, District 5, HCMC, Vietnam. Telephone No.: +842838358411 E-mail: caotran_lephuong@yahoo.com Orchid Identifier: https://orcid.org/0000-0003-3538-6604
	Dr Dinesh Neupane (DN) PhD Research Associate, Welch Center for Prevention, Epidemiology and Clinical Research, Department of Epidemiology, Johns Hopkins Bloomberg School of Public Health Baltimore, Maryland, Baltimore, USA. E-mail: dneupan2@jh.edu Orchid Identifier: https://orcid.org/0000-0002-1501-2990
Corresponding Author	Mr. Shuvarthi Bhattacharjee (SB) Early Stage Researcher, School of Health Sciences, University of Brighton (Falmer Campus), Brighton, BN19PH.UK. Telephone No.: +441273644025 E-mail: S.Bhattacharjee@brighton.ac.uk Orchid Identifier: https://orcid.org/0000-0002-6079-5365
CONTRIBUTIONS (3b) (Authors)	SB is the guarantor of the review. SB & NY conceptualized the study. A search strategy will be created and run on relevant platforms by SB. All search results will be retrieved by SB from appropriate databases, trial registry (for details of primary studies in reviews) & grey literature. CLP & NY will validate the search strategy prior to implementation and check the retrieved search results. SB, NY and CLP will screen articles and any discrepancies for final inclusion of Full- text articles will be

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	resolved by DN (fourth researcher). All authors will be involved in data extraction, quality appraisal, data synthesis and formulation of the final draft.
AMENDMENTS (4)	Important protocol amendments post registration will be recorded, updated in PROSPERO and included in knowledge dissemination.

SUPPORT	
Sources (5a)	No Financial support was received for this study. Authors are academic students or staff in universities & civil society organizations. Access to Healthcare Databases Advanced Search (HDAS) will be provided by NHS Openathens, UK for searching of databases. Access to Web of Science will be provided by the University of Brighton (UK). Data Management support will be sought from COVIDENCE® systematic review software.
Sponsor (5b)/Funding	This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.
Role of sponsor/funder (5c)	Not Applicable.

Methodology for this Protocol
We followed the PRISMA-P guidelines which recommend items to address in a systematic review protocol.

Abstract

Introduction: Ischemic Heart Disease (IHD) is one of the leading causes of death & disease burden in India affecting all age groups. To reduce the deaths and tackle the burden of existing IHD the government approach has been mostly through the National Health Policy (2017) and National Program for Prevention & Control of Diabetes, Diabetes, Cardiovascular diseases and Stroke (NPCDCS). This paper offers a protocol for the systematic review of studies exploring the factors influencing service readiness of the public health system of India to tackle the burden of IHD.

Methods and Analysis: : Electronic databases of EMBASE (Ovid), AMED(Ovid), HMIC(Ovid), BNI(ProQuest), CINAHL(EBSCO), EMCARE(Ovid), PsycINFO(ProQuest), Medline/PubMed and Web of Science (Clarivate Analytics) will be searched till 1st October, 2020 for primary studies. Grey literature will be accessed through OpenGrey, TRIP Medical, WHO Database, MoHFW Website, OGD Platform India and Google Scholar (between 2010-2020). Primary studies meeting the eligibility criteria & grey literature published in English between 2010 to 2020 will be included. Data will be analysed through a

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

conceptual framework and primary outcome will constitute both quantitative and qualitative data. The quality of included studies will be assessed based on study design. Data will be managed on the COVIDENCE® platform. All authors will be involved in data extraction, quality appraisal, data synthesis and formulation of the final draft.

Ethics and dissemination: This study being a systematic review does not involve any clinical trial, primary data collection or empirical study involving humans or animals. Therefore, no ethical permissions were sought by reviewers.

Registration details: This systematic review protocol is registered in the International Prospective register of Systematic Reviews (PROSPERO) with number CRD 42020219490.

Link: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=219490

Strengths and limitations of this study

- To our knowledge, this is the first systematic review on the factors influencing readiness of the public health system of India to tackle the burden of Ischemic Heart Disease.
- The study can inform policymakers, academicians, NGOs and healthcare administrators with understanding of the existing gaps in the health system.
- This study can provide options to optimise existing health system resources of India to extend better quality of care for the patients.
- Our use of a deductive approach with a conceptual framework for data analysis may exclude some findings which might be a limitation.
- India being a vast country with a decentralised system of health governance might have varied levels of service readiness across states making it difficult to generalise study findings for all contexts.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

INTRODUCTION

6. Rationale

Ischemic Heart Disease (IHD) is the leading cause of deaths and disease burden in the adults of India. In 2019, IHD alone claimed approximately 1.5 million deaths in India with a mortality rate of 109.23 deaths per 100000 populations**.[1]** Between 2009 and 2019, deaths due to IHD increased significantly by 29.8% in India and in the year 2019 contributed to 7.79% of the total Disability Adjusted Life Years (DALYs). IHD affected all age groups including the working population of the country, leading to their premature deaths. High burden of disease & deaths due to IHD can have significant socio-economic implications for India. Specifically, the working population of India who are earning members within households may suffer premature deaths, leading to loss of individual lives and resulting in economic instability for their dependents. Moreover, healthcare expenditure is majorly met through out of pocket (OOP) expenses, presently at 63% of current health expenditure (CHE) of India, pushing households towards financial impoverishment.**[2]** In such a situation, tackling the disease burden and reducing deaths due to IHD seems necessary to avert loss of individual lives and prevent possible financial impoverishment of their households.

In order to reduce the deaths and tackle the burden of IHD in India the government approach has been mostly through context centric tactics of tackling the Non-Communicable Diseases (NCDs) through nationwide health sector policy & programs.**[3]** In agreement with the Global Action Plan for Prevention & Control of NCDs 2013-2020, endorsed by the World Health Assembly (WHA) in 2013, India has adopted 25 indicators and 9 voluntary goals to reduce 25% of premature deaths due to NCDs by 2025.**[4,5]** Premature mortality associated with IHD is indicated as unconditional probability of dying due to IHD between the age bands of 30-70 years. In India, the translation of this global action plan at national level has occurred through the National Health Policy (NHP-2017) and the policy vision is being implemented across states through multiple approaches by the National Program for Prevention & Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS).**[6,7]** Additionally, apart from this, India's flagship program on universal health coverage (UHC), Ayushman Bharat, has plans to convert 150000 Primary Health Centres (PHCs) into Health & Wellness Centres (HWCs) to strengthen primary care services that can prevent & treat NCDs including IHD.**[8]** To support the cause, the NHP-2017 planned to raise the General Government Health Expenditure (GGHE) to 2.5% of the GDP from the usual government allocation of 1.2% of the GDP in previous years. Moreover, NHP-2017 has suggested NPCDCS to strengthen secondary & tertiary care services, along with primary care NCD interventions, such as 'Best Buys'.**[9]**

However, while the plans and commitments have been put in place, the challenges of the existing disease burden of IHD, upscaling premature deaths due to IHD, overwhelmed public healthcare system, scarcity of health workers and the poor co-operation between states and the central government in India's decentralized

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

health administrative structure appear to be detrimental towards tackling the disease burden of IHD.**[10-12]** Therefore, it becomes necessary to explore the factors influencing readiness of the public health system of India in order to tackle IHD based upon the commitments made through the NHP-2017 & other relevant national programs to tackle the increasing number of cases & deaths due to IHD. Readiness here is defined as the cumulative availability of components required to provide services.**[13]**

In a nutshell, this is an attempt to explore whether the existing public health system is capable of delivering the government commitments on national health policy and national NCD program to the people of India, through the existing public health system infrastructure, in order to tackle the mortality & morbidity due to IHD.

Our preliminary search on this topic in Cochrane Database of Systematic Reviews, PROSPERO, JBI Evidence Synthesis, CRD/DARE Database and Campbell Collaboration revealed no systematic reviews, scoping reviews, rapid reviews or published protocols related to India concerning readiness of health system. However, there were reviews focussed on health system components for Low and Middle Income Countries (LMIC) settings, acute care settings concerning LMIC countries, monitoring of cardiac patients and risk factors associated with IHD in Asian Indians (included those of Indian origin living outside India) which bear no similarity to our study.**[14-18]** Moreover, data from such studies considered LMIC context in a totality which made it difficult to find any specific data for India. The COVID-19 pandemic in India is already witnessing a scarcity of oxygen supply and overwhelmed tertiary care sector. **[19]** Further, health system issues like dual practice, poor skill mix and inadequate number of healthcare staff in the public sector has further constrained the existing care system.**[20]** In this scenario people with IHD or at risk with IHD might overstrain the existing care system especially with cases requiring tertiary care services.

7. Objectives

To identify the health system factors that contributes to facility readiness of the public healthcare system of India in tackling the burden of IHD. To achieve the stated objective a systematic review to analyse the existing peer reviewed studies and grey literature is proposed to be undertaken. The analysis will be guided through a conceptual framework.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

METHODS

8.Eligibility Criteria

i) Population, or participants and conditions of interest

Studies involving Indian people living with or at risk of IHD. Primary studies related to such population type published in peer reviewed journals and grey literature in English published between 2010 and 2020 will be included.

ii) Interventions or exposures

Relevant primary studies in peer reviewed journals & grey literature on national programs of NCD, National Health Policies (NHP-2012 & NHP-2017), interventions concerned with both prevention & control of Ischemic Heart Disease in India will be included if published in English between 2010 to 2020.

iii) Comparisons or control groups None.

iv) Outcomes of interest

Data of both quantitative & qualitative nature related to health system preparedness or readiness to prevent & tackle the burden of Ischemic Heart Disease in India.

v) Setting

Primary, Secondary & Tertiary Care settings in Public Health System of India.

vi) Study Designs

All forms of study design of both quantitative & qualitative nature will be included.

vii) Time Frame: 2010-2020. This timeframe has been chosen considering that the NPCDCS was launched by the Ministry of Health & Family Welfare, Government of India, in 2010. **[21]**

viii) Reporting Characteristics:

Years Considered: 2010 to 2020; Language: English; Publication Status: Peer Reviewed.

9. Information Sources: Details of Information sources has been outlined in ***Supplementary Material A: Information Sources & Representative Search Strategy.***

10. Search Strategy: Electronic databases of EMBASE (Ovid), AMED(Ovid), HMIC(Ovid), BNI(ProQuest), CINAHL(EBSCO), EMCARE(Ovid), PsycINFO(ProQuest), Medline/PubMed and Web of Science (Clarivate Analytics) will be searched with use of appropriate Boolean operators (AND, OR & NOT) till 2020. No filters or limits will be used at this stage. All electronic databases (except Medline/PubMed and Web of Science) will be accessed through the Healthcare Databases for NHS in England (HDAS) of the NHS Openathens, UK. HDAS mirrors search results of platforms like Ovid, EBSCO and ProQuest to retrieve results from these databases.

Trial registers like PROSPERO, Cochrane Database of Systematic Reviews, Campbell Collaboration and CRD/DARE Database will be searched to avoid duplication of existing knowledge on the topic and identify relevant primary studies from available systematic reviews.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Moreover, citation tracking, footnote chasing, bibliography scan and citation alerts (on web of science) will be conducted for a comprehensive search.**[22]** A combination of free texts and corresponding synonyms of terms comprising but not limited to "Ischemic Heart Disease" AND "Health System" AND "Readiness" AND "India" will be made for the initial search and appropriate preidentified MeSH Terms ('Emtree Terms' for Embase) in PubMed/Medline will be included to make a comprehensive search strategy across the electronic databases (Table 1 added as a representative example). Thesaurus terms appropriate to the databases will be used to account for the variation in such terms across databases.

Grey literature will be accessed through OpenGrey, TRIP Medical, WHO Database, MoHFW Website, OGD Platform India and Google Scholar (between 2010-2020).

Detailed list of information sources & representative search result for this study are given in ***'Supplementary Material A: Information Sources & Representative Search Strategy'.***

STUDY RECORDS

11a. Data management

COVIDENCE® systematic review software (available at : www.covidence.org) will be employed to manage the studies retrieved and initiate the process of systematic review. On this software duplicate articles will be rejected in order to gather a set of unique articles for this study.**[23]** All article screening activities, selection of final set of articles for inclusion in the study after examination of full texts of eligible studies, data extraction and risk of bias assessment will be carried out on COVIDENCE®. References will be managed and stored on Endnote®. All forms of article selection, data extraction, quality appraisal, risk of bias assessment will be performed and a 'PRISMA flow chart', based on PRISMA guidelines for systematic review, will be created on the COVIDENCE® platform.

11b. Selection process

Initially, all duplicate articles will be checked by three reviewers (SB, NY, CLP) individually through a double blinded process to assure a rigorous review approach and rejected before identifying a unique set of articles for this study on the COVIDENCE® platform.

Afterwards, the screening of title and abstracts of potential articles will be conducted by the same reviewers in a similar double blinded manner to assure rigor of the study.**[24]** In case of any conflict related to the inclusion of an article, it will be temporarily included in the next step for further evaluation.

In the next step, full-text articles will be examined based on the inclusion and exclusion criteria before final inclusion in the study. Reasons for the inclusion and exclusion of articles will be clearly identified and documented by each reviewer independently on ***Supplementary Material B*** (Form for Screening of full text articles based on inclusion & exclusion criteria with documented justification). Assuring three reviewers will allow for a consensus-based conflict resolution in case there is uncertainty raised about inclusion of an

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

article for the study. Moreover, the fourth researcher will also be consulted before finalising the final set of included articles for this study.

11c. Data Collection Process

Data will be extracted based on ***Supplementary Material C*** (Data Extraction Form for Included Studies) by three reviewers (SB, NY & CLP) independently in a blind manner for each included study. SB will randomly select half of the included studies which will be given to the fourth researcher to extract data independently. This will ensure rigor of the study and a consensus-based approach for approving the final content of the data extraction form. Any disagreements will be resolved through mediation with the fourth researcher. The data extraction form will also offer content on the risk of bias and quality assessment of included studies. As the final data synthesis will involve using a conceptual framework, all relevant quantitative data will be converted into meaningful qualitative data for analysis and interpretation.

12. Data Items

The descriptive data items that will be collected constitute : (i) Characteristics of the study including country & year of publication, research aim & participant characteristics; (ii) Methodological characteristics including study design, data/information collection methods and data analysis; (iii) Results associated with readiness of the public health system to tackle the burden of ischemic heart disease along with concluding remarks from the article. The data collection items will also include risk of bias and quality assessment of included individual articles based on appropriate tools. The proposed tools and process for assessment of risk of bias & quality assessment of eligible studies is further detailed in the section pertaining to 'risk of bias in individual studies' within this protocol. For this study we wish to create simplification of quantitative data by transforming them to meaningful qualitative units to accommodate them into the conceptual framework opted for our study. Moreover, only relevant data will be extracted based on the needs of the study.

We will extract data from those studies concerned with primary care facilities where facility assessment can provide data pertaining to facility readiness even if readiness concerning ischemic heart disease is not mentioned. This is because treatment of Ischemic Heart Disease and its risk factors are integrated at the primary care level along with treatment of other communicable and non-communicable diseases.**[25]**

13. Outcomes and prioritization

Outcome of this study will constitute both quantitative and qualitative data retrieved from the included articles to inform health system factors that influence the readiness of India's public health system to tackle the burden of Ischemic Heart Disease.

This will include data at all levels of care that informs health service delivery, health workforce, health information system, access to essential medicines, health financing and health system governance concerning the present topic.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

In terms of quantitative data, data from facility readiness assessment, health facility assessment, WHO Service Availability & Readiness Assessment (SARA) data, surveys, baseline study of health facilities and monitoring & evaluation data of health facilities along with data from the national NCD program of India will serve to provide information on the outcome.

In terms of qualitative studies, data from in-depth interviews, focus group discussions, researcher observations of facilities and data from the national health policies will serve to understand the outcomes better.

14. Risk of bias in individual studies

All eligible studies at the outcome stage are expected to have methodological variation in approach and therefore, we will assess their quality & risk of bias at this stage. In order to minimize the risk of bias and evaluate the quality of included studies the Cochrane collaboration tool for assessing the risk of bias of Randomized Control Trials (RCTs), Newcastle-Ottawa Scale for cohort & case-control studies, NIH Quality Assessment Tool for Cross-Sectional Studies and Critical Appraisal Skills Program (CASP) tool for qualitative studies will be used based on the methodological design of the individual eligible study. **[26-29]** We will not reject any study based on assessed quality only the information will be documented in the data extraction and synthesis stage. The risk of bias assessment and quality appraisal will be independently performed by each reviewer before converging to a mutually agreed decision. Any discrepancies will be mediated through the fourth researcher.

National Health Policies of India, national NCD program guidelines and raw primary data from the OGD Platform cannot be assessed for risk of bias and quality appraisal. They will be excluded from this process of evaluation.

15. DATA SYNTHESIS

15 (a &b) Data Synthesis

The extracted data will be analysed through a conceptual framework, the 'WHO Framework for Action', followed by summarization of findings to answer the research questions.**[30]** We do not expect to conduct a meta-analysis as we do not anticipate many articles in which meta-analysis may be appropriate. Additionally, the inclusion of qualitative studies will likely lead to a heterogeneous outcome which is not suitable for meta-analysis. Moreover, we will be transforming the quantitative data to meaningful qualitative units for assimilation into the conceptual framework for our study. Data analysis through the proposed conceptual framework rather serves as a conceptual lens for this study. Therefore, we will be deductive in our approach assuming all health system readiness issues are inclusive within this conceptual framework. The conceptual framework was chosen through mediation with all reviewers and moreover, straight forward demarcation of health system components allows for clear interpretation.**[31]** We also considered other

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

conceptual frameworks; however, they were either overly complex for the purposes of this study or considered factors beyond the health system that were not relevant for the present study.**[32-36]**

15 (c) Sub-Group Analysis

We anticipate that sub-group analysis might be performed from the extracted data by using the components of 'Health System Strengthening' (HSS) building blocks from the proposed conceptual framework.

15 (d) Data Summary

Data will be summarized thematically based on 'Health System Strengthening' (HSS) building blocks from the proposed conceptual framework.

16. Meta-Bias(es)

We do not plan to conduct any assessment of meta-bias(es) as the expected studies included for this systematic review might be of varied study design making it difficult to assess such bias such as 'publication bias across studies' or 'selective reporting within studies'. **[37-38]**

In this context, our approach to this review might be influenced by 'language bias' as we are including studies published only in English. We plan to mention this aspect as our study limitation in the final review. Further, performing citation tracking can lead us to a "biased sample of studies". **[39]** We plan to mitigate this by involving multiple authors in the screening, data extraction and data analysis process. Moreover, using a particular conceptual framework with a deductive research approach might lead us to a selective reporting of outcomes. We plan to avoid this through collective discussions between all the researchers post data extraction stage.

17. Confidence in cumulative evidence

At present we do not have plans to assess the confidence in cumulative evidence as this study does not report effectiveness of clinical outcome(s).

18. Presenting or reporting the results

The proposed systematic review will transparently represent the entire screening process and offer information on the finally included studies through a flow diagram in accordance with the PRISMA-P guidelines**.[40]** Data collection will be through a data extraction form with outcomes to extract data related to health system readiness. Data will be summarized in tables with details of study characteristics, methodological characteristics and individual study results. We will also provide data of included and excluded studies with documented justification of their inclusion or exclusion.

19. Patient and public involvement

The present study is a systematic review not an empirical study and therefore, there is no direct involvement of patient or public in this study.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

20. Ethics & Dissemination

20 (a) Ethical Issues

This study being a systematic review does not involve any clinical trial, primary data collection or empirical study involving humans or animals. Therefore, no ethical permissions were sought by reviewers.

20 (b) Publication Plan

We plan to publish the preliminary results of this review in conferences and proceed on to publish the final article in peer reviewed journal once completed. We hope to publish both the protocol & systematic review in open access journals to benefit academic communities, policymakers and other public health stakeholders with the ambition to improve the readiness of the health to tackle the burden of Ischemic Heart Disease in India and elsewhere.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

References:

1. Institute for Health Metrics and Evaluation (IHME). Findings from the Global Burden of Disease Study 2019. Seattle, WA: IHME, 2021. Available from: https://vizhub.healthdata.org/gbd-compare/
2. World Health Organization (WHO). Global health expenditure database. Last updated August 20, 2020. http://www.who.int/health-accounts/ghed/en/ (accessed Sept. 18, 2020).
3. Ministry of Health & Family Welfare (MoHFW, India). National Health Portal (2019, November 5). Non-communicable Diseases. Retrieved from https://www.nhp.gov.in/healthlyliving/ncd2019 (accessed Sept 18, 2020).
4. WHO. Global action plan for the prevention and control of noncommunicable diseases 2013-2020. World Health Organization. 2013 Mar 29:102.
5. Ministry of Health & Family Welfare (MoHFW, India). National Action Plan and Monitoring Framework for prevention & control of Noncommunicable Diseases (NCDs). Retrieved from https://www.iccp-portal.org/system/files/plans/India%20-%20National_Action_Plan_and_Monitoring_Framework_Prevention_NCD_2013.pdf (accessed on Sept 18, 2020).
6. Ministry of Health & Family Welfare (MoHFW, India). National Health Policy; 2017. Available from: http://www.cdsco.nic.in/writereaddata/National-Health-Policy.pdf (accessed on Sept. 18, 2020).
7. Ministry of Health & Family Welfare (MoHFW, India). Directorate General of Health Services (DGHS). National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke. Retrieved September 18, 2020, from https://dghs.gov.in/content/1363_3_NationalProgrammePreventionControl.aspx
8. Gopichandran V. Ayushman Bharat National Health Protection Scheme: An Ethical Analysis. Asian Bioethics Review 2019;11:69-80. doi:10.1007/s41649-019-00083-5
9. World Health Organization. Tackling NCDs: 'best buys' and other recommended interventions for the prevention and control of noncommunicable diseases. World Health Organization; 2017.
10. Narain JP. Public health challenges in India: seizing the opportunities. Indian journal of community medicine: official publication of Indian Association of Preventive & Social Medicine. 2016 Apr;41(2):85.
11. Tiwari R, Negandhi H, Zodpey S. Health Management Workforce for India in 2030. Frontiers in Public Health 2018;6. doi:10.3389/fpubh.2018.00227

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12. Lakshminarayanan S. Role of government in public health: Current scenario in India and future scope. Journal of Family and Community Medicine 2011;18:26. doi:10.4103/1319-1683.78635
13. Abazinab S, Woldie M, Alaro T. Readiness of health centers and primary hospitals for the implementation of proposed health insurance schemes in Southwest Ethiopia. Ethiopian Journal of Health Sciences 2016;26:449. doi:10.4314/ejhs.v26i5.6
14. Marin T, Walsh S, May N et al. Screening for depression and anxiety among patients with acute coronary syndrome in acute care settings: a scoping review. JBI Evidence Synthesis 2020;18:1932-1969. doi:10.11124/jbisrir-d-19-00316
15. Likka M, Handalo D, Weldsilase Y et al. The effect of community-based health insurance schemes on utilization of healthcare services in low- and middle-income countries: a systematic review protocol of quantitative evidence. JBI Database of Systematic Reviews and Implementation Reports 2018;16:653-661. doi:10.11124/jbisrir-2017-003381
16. Teferra M, Ramos J, Kourbelis C et al. Electronic textile-based electrocardiogram monitoring in cardiac patients. JBI Database of Systematic Reviews and Implementation Reports 2019;17:1958-1998. doi:10.11124/jbisrir-2017-003989
17. Ninan A, Acosta J, Kulesza T et al. A systematic review of the effectiveness of preventive measures for coronary artery disease in Asian Indians. JBI Library of Systematic Reviews 2011;9:1-17. doi:10.11124/jbisrir-2011-314
18. Ninan A, Acosta J, Kulesza T et al. A systematic review on the risk factors of coronary artery disease found in Asian Indians. JBI Library of Systematic Reviews 2011;9:1-14. doi:10.11124/jbisrir-2011-455
19. Kumar P, Kumar M. Management of potential ventilator shortage in India in view of on-going COVID-19 pandemic. Indian Journal of Anaesthesia 2020;64:151. doi:10.4103/ija.ija_342_20
20. Iyengar KP, Jain VK, Vaishya R. Current situation with doctors and healthcare workers during COVID-19 pandemic in India. Postgraduate Medical Journal Published Online First: 19 August 2020. doi: 10.1136/postgradmedj-2020-138496
21. Arokiasamy P. India's escalating burden of non-communicable diseases. The lancet global health. 2018 Dec 1;6(12):e1262-3.
22. Wright K, Golder S, Rodriguez-Lopez R. Citation searching: a systematic review case study of multiple risk behaviour interventions. BMC Medical Research Methodology 2014;14. doi:10.1186/1471-2288-14-73
23. Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia. Available at www.covidence.org .2021.
24. Ayala Quintanilla B, Taft A, McDonald S et al. Social determinants and maternal exposure to intimate partner violence of obstetric patients with severe maternal morbidity in the intensive care unit: a systematic review protocol. BMJ Open 2016;6:e013270. doi:10.1136/bmjopen-2016-013270

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

25. Ministry of Health & Family Welfare, Government of India. National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Disease and Stroke: Operational Guidelines. Strategy. 2010.
26. Higgins J, Altman D, Gotzsche P et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. BMJ 2011;343:d5928-d5928. doi:10.1136/bmj.d5928
27. Peterson J, Welch V, Losos M, Tugwell PJ. The Newcastle-Ottawa scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Ottawa: Ottawa Hospital Research Institute. 2011.
28. National Institutes of Health (NIH). Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies. 2014. Available online at: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools (Accessed November 19, 2020).
29. Critical Appraisal Skills Programme. (CASP) qualitative checklist, 2018, https://casp-uk.net/wp-content/uploads/2018/01/CASP-Qualitative-Checklist-2018.pdf (accessed 18 Sept. 2020)
30. World Health Organization. Everybody's business--strengthening health systems to improve health outcomes: WHO's framework for action. 2007.
31. Mounier-Jack S, Griffiths U, Closser S et al. Measuring the health systems impact of disease control programmes: a critical reflection on the WHO building blocks framework. BMC Public Health 2014;14. doi:10.1186/1471-2458-14-278
32. Roberts, M.J., W.C. Hsiao, P. Berman, and M.R Reich. 2003. Getting Health Reform Right. New York: Oxford University Press, page 27
33. De Savigny, D. and Adam, T. (eds.) 2009, Systems thinking for health systems strengthening, World Health Organization, Geneva.
34. Shakarishvili G, Lansang M, Mitta V et al. Health systems strengthening: a common classification and framework for investment analysis. Health Policy and Planning 2010;26:316-326. doi:10.1093/heapol/czq053
35. Josefien van Olmen, Bart Criel, Wim Van Damme, Bruno Marchal, Sara Van Belle, Monique Van Dormael, Tom Hoerée, Marjan Pirard and Guy Kegels (2010). Analysing health systems to make them stronger.ITG Press, Antwerpen, Belgium.
36. Murray, C. J., & Frenk, J. (2000). A framework for assessing the performance of health systems. Bulletin of the world Health Organization, 78, 717-731.
37. Dalton J, Bolen S, Mascha E. Publication Bias. Anaesthesia & Analgesia 2016;123:812-813. doi:10.1213/ane.0000000000001596
38. Reid E, Tejani A, Huan L et al. Managing the incidence of selective reporting bias: a survey of Cochrane review groups. Systematic Reviews 2015;4. doi:10.1186/s13643-015-0070-y
39. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.2 (updated February 2021). Cochrane, 2021. Available from www.training.cochrane.org/handbook.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

40. Moher D, Liberati A, Tetzlaff J et al. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Medicine 2009;6:e1000097. doi:10.1371/journal.pmed.1000097

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Material A: Information Sources & Representative Search Strategy			
(i) Electronic Databases:	**Database**	**Accessed From**	**Planned Dates of Coverage**
	Pubmed	NIH Website	From 2010 till 01.10.2020
	Embase	NHS HDAS Platform	From 2010 till 01.10.2020
	EMCARE	NHS HDAS Platform	From 2010 till 01.10.2020
	HMIC	NHS HDAS Platform	From 2010 till 01.10.2020
	BNI	NHS HDAS Platform	From 2010 till 01.10.2020
	CINAHL	NHS HDAS Platform	From 2010 till 01.10.2020
	AMED	NHS HDAS Platform	From 2010 till 01.10.2020
	PsycINFO	NHS HDAS Platform	From 2010 till 01.10.2020
	Web of Science	Clarivate Analytics	From 2010 till 01.10.2020
(ii)Trial Registers	PROSPERO	CRD Database	From 2010 till 01.10.2020
	Cochrane Database of Systematic Reviews	Cochrane Library	From 2010 till 01.10.2020
	Campbell Collaboration	Cambell Collaboration Website.	From 2010 till 01.10.2020

For peer review only

	JBI Evidence Synthesis	Joanna Briggs Institute Website.	From 2010 till 01.10.2020
	CRD/DARE Database	Centre for Reviews and Dissemination Website.	From 2010 till 01.10.2020.
(iii) Contact with study authors	Researchgate	Researchgate Website	As per necessity to request for full text articles based on articles published till 01.10.2020.
(iv) Grey Literature Sources	OpenGrey	INIST-CNRS managed Website	01.01.2010 to 01.10.2020
	TRIP Medical	TRIP Database Website	01.01.2010 to 01.10.2020
	WHO Database	World Health Organization Website	From 2010 till 01.10.2020.
	MoHFW	Ministry of Health & Family Welfare Website.	From 2010 till 01.10.2020.
	OGD Platform India	Government of India Website on Open Government Data.	From 2010 till 01.10.2020.
	Google Scholar	Google Scholar Website (only first 10 pages)	From 2010 till 01.10.2020.

For peer review only

Table 1		
A representative search for CINAHL database performed on NHS Openathens HDAS Platform on 08th August 2020. No Filters or Limits were applied.		
Sl. No.	**Search Terms**	**Results**
1	(India*).ti,ab OR (South Asia*).ti,ab OR (Asia*).ti,ab OR (LMIC).ti,ab OR (LIC).ti,ab OR (Low Income Countr*).ti,ab OR (Low Middle Income Countr*).ti,ab OR (SAARC).ti,ab OR (ASEAN).ti,ab	85642
2	(NCD*).ti,ab OR (non-communicable diseas*).ti,ab OR (chronic disease* OR myocardial ischemia).ti,ab OR (cardio?vascular diseas*).ti,ab OR (ischemic heart diseas* OR heart disease*).ti,ab OR (vascular disease*).ti,ab OR (Diabetes).ti,ab OR (Hypertension).ti,ab OR ("High Blood Pressure").ti,ab OR (Cancer).ti,ab OR (COPD).ti,ab OR (Chronic Obstructive Pulmonary Disease).ti,ab OR (Coronary Artery Diseas*).ti,ab OR (Coronary Heart Diseas*).ti,ab	789662
3	(health system* OR program*).ti,ab OR (health?care system*).ti,ab OR (health planning* OR health service utiliz*).ti,ab OR (health?care planning*).ti,ab OR (health facilit*).ti,ab OR (National Health Policy).ti,ab OR (National Health Program*).ti,ab OR (NPCDCS).ti,ab OR "primary care" OR "primary healthcare" OR "secondary care" OR healthcare OR "tertiary care" OR hospital*	1503977
4	(prepare* OR health system prepared*).ti,ab OR (ready).ti,ab OR (readiness).ti,ab OR (facility? assess*).ti,ab OR (readiness).ti,ab	77767
5	exp INDIA/ OR exp "ASIA, WESTERN"/	128168
6	exp ASIANS/	32187
7	exp "DEVELOPING COUNTRIES"/	18662
8	exp "LOW- AND MIDDLE INCOME COUNTRIES"/	1361
9	(5 OR 6 OR 7 OR 8)	176910

For peer review only

10	(1 OR 9)	225727
11	exp "CARDIOVASCULAR DISEASES"/ OR exp "HEART DISEASES"/ OR exp "VASCULAR DISEASES"/	584557
12	exp "CORONARY DISEASE"/	54503
13	exp "MYOCARDIAL ISCHEMIA"/ OR exp "ACUTE CORONARY SYNDROME"/	101965
14	exp "CHRONIC DISEASE"/ OR exp "NONCOMMUNICABLE DISEASES"/	64096
15	exp "DIABETES MELLITUS"/	164718
16	exp HYPERTENSION/	78476
17	exp "LUNG DISEASES, OBSTRUCTIVE"/	60675
18	exp NEOPLASMS/	544583
19	(11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18)	1311975
20	(2 OR 19)	1554463
21	(10 AND 20)	50731
22	exp "HEALTH AND WELFARE PLANNING"/ OR exp "HEALTH FACILITY PLANNING"/	171562
23	exp "HEALTH CARE DELIVERY"/	338701
24	exp "NATIONAL HEALTH PROGRAMS"/ OR exp "HEALTH RESOURCE UTILIZATION"/ OR exp "HEALTH SERVICES ACCESSIBILITY"/ OR exp "MANAGED CARE PROGRAMS"/ OR exp "PRIMARY HEALTH CARE"/ OR exp "SECONDARY HEALTH CARE"/ OR exp "TERTIARY HEALTH CARE"/	237596
25	exp "PUBLIC HEALTH ADMINISTRATION"/ OR exp "PROGRAM DEVELOPMENT"/	88044
26	exp "PUBLIC POLICY"/ OR exp "HEALTH POLICY"/	106774
27	exp "HEALTH POLICY STUDIES"/ OR exp "POLICY STUDIES"/	827
28	(22 OR 23 OR 24 OR 25 OR 26 OR 27)	526561
29	(3 OR 28)	1709631
30	(21 AND 29)	20354

For peer review only

31	(4 AND 30)	423
32	(4 AND 30) [DT 2010-2020] [Peer reviewed] [Languages eng]	326

For peer review only

Supplementary Material B: Form for Screening of full text articles based on inclusion & exclusion criteria with documented

Basic Inclusion Criteria								SB		CLP		NY		
Sl. No.	Title	Is this peer reviewed?	Is this paper in English?	Is this a primary study?	Published between 2000 to 2020?	Does it concern India?	Article related to the topic?	Reasons for Inclusion	Reasons for Exclusion	Reasons for Inclusion	Reasons for Exclusion	Reasons for Inclusion	Reasons for Exclusion	Final Decision (Yes/No)
1														
2														
3														
4														
5														
6														
7														
8														
9														
10														

For peer review only

Supplementary Material C: Data Extraction Form for Included Studies									
Sl. No.	Title	Author	Year & Country	Research Aims	Study Design	Participants	Methods	Results	Conclusions
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									

For peer review only

Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

For peer review only

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	1

Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	1
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1-2
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	2
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	3
Support			
Sources	#5a	Indicate sources of financial or other support for the review	3
Sponsor	#5b	Provide name for the review funder and / or sponsor	3
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	3
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is	5-6

For peer review only

		already known	
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7-8
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Study records - data collection	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any	9

For peer review only

process		processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	9
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	9-10
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	10
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	10-11
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	N/A
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	11
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	11
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within	11

For peer review only

		studies)	
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	11

The PRISMA-P checklist is distributed under the terms of the Creative Commons Attribution License CC-BY 4.0. This checklist was completed on 29. November 2020 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

For peer review only

BMJ Open

Factors influencing the readiness to tackle the burden of ischemic heart disease in India: a systematic review protocol.

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-047464.R2
Article Type:	Protocol
Date Submitted by the Author:	07-Jun-2021
Complete List of Authors:	Bhattacharjee, Shuvarthi; University of Brighton, School of Health Sciences Yaghmaei , Nima ; HumERA Consulting, Phuong, Cao; Ho Chi Minh City University of Medicine and Pharmacy, Department of Public Health Neupane, Dinesh; Johns Hopkins University Bloomberg School of Public Health, Epidemiology
Primary Subject Heading:	Public health
Secondary Subject Heading:	Health policy, Health services research, Global health
Keywords:	Ischaemic heart disease < CARDIOLOGY, Health policy < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, HEALTH SERVICES ADMINISTRATION & MANAGEMENT, PUBLIC HEALTH, Cardiac Epidemiology < CARDIOLOGY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

ADMINISTRATIVE INFORMATION

1. Title of the review	**Factors influencing the readiness to tackle the burden of ischemic heart disease in India: a systematic review protocol.**
2. Registration	This systematic review protocol is registered in the International Prospective Register of Systematic Reviews (PROSPERO) with number CRD 42020219490 following method guidelines from Cochrane handbook of systematic reviews of interventions to conduct of this systematic review, reported in line with the PRISMA statement. Link: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=219490
3. Keywords	Humans; India; Myocardial Ischemia; Systematic Review; Health Services; Noncommunicable Diseases;
4. Word Count	3688 words excluding abstract & references.
5. Conflicts of Interest	The Author(s) declare(s) that there is no conflict of interest.
6. Acknowledgements	None.
7.Data Statement	No dataset is available for this systematic review yet. However, preliminary search result can be found in the following link: https://www.crd.york.ac.uk/PROSPEROFILES/219490_STRATEGY_20201106.pdf
8. Review Status	Ongoing. At present undergoing searches across electronic databases.

AUTHORS

CONTACT(3a)	
	Mr. Shuvarthi Bhattacharjee (SB) M.Sc. (Public Health) Early Stage Researcher, School of Health Sciences, University of Brighton (Falmer Campus), Brighton, BN19PH.UK. Telephone No.: +441273644025 E-mail: S.Bhattacharjee@brighton.ac.uk Orchid Identifier: https://orcid.org/0000-0002-6079-5365
	Mr. Nima Yaghmaei (NY) MIH Independent Consultant, HumERA Consulting, 115, Esgore Drive, Toronto, Ontario, Canada M5M3S1

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	Telephone No.: + 32498068096 E-mail: nimayagh@gmail.com Orchid Identifier: https://orcid.org/0000-0001-7740-8037
	Ms. Cao Tran Le Phuong (CLP) M.Sc. (Public Health) Researcher, Public Health Department, University of Medicine & Pharmacy, 217, Hong Bang Street, Ward 11, District 5, HCMC, Vietnam. Telephone No.: +842838358411 E-mail: caotran_lephuong@yahoo.com Orchid Identifier: https://orcid.org/0000-0003-3538-6604
	Dr Dinesh Neupane (DN) PhD Research Associate, Welch Center for Prevention, Epidemiology and Clinical Research, Department of Epidemiology, Johns Hopkins Bloomberg School of Public Health Baltimore, Maryland, Baltimore, USA. E-mail: dneupan2@jh.edu Orchid Identifier: https://orcid.org/0000-0002-1501-2990
Corresponding Author	Mr. Shuvarthi Bhattacharjee (SB) Early Stage Researcher, School of Health Sciences, University of Brighton (Falmer Campus), Brighton, BN19PH.UK. Telephone No.: +441273644025 E-mail: S.Bhattacharjee@brighton.ac.uk Orchid Identifier: https://orcid.org/0000-0002-6079-5365
CONTRIBUTIONS (3b) (Authors)	SB is the guarantor of the review. SB & NY conceptualized the study. A search strategy will be created and run on relevant platforms by SB. All search results will be retrieved by SB from appropriate databases, trial registry (for details of primary studies in reviews) & grey literature. CLP & NY will validate the search strategy prior to implementation and check the retrieved search results. SB, NY and CLP will screen articles and any discrepancies for final inclusion of Full- text articles will be

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	resolved by DN (fourth researcher). All authors will be involved in data extraction, quality appraisal, data synthesis and formulation of the final draft.
AMENDMENTS (4)	Important protocol amendments post registration will be recorded, updated in PROSPERO and included in knowledge dissemination.

SUPPORT	
Sources (5a)	No Financial support was received for this study. Authors are academic students or staff in universities & civil society organizations. Access to Healthcare Databases Advanced Search (HDAS) will be provided by NHS Openathens, UK for searching of databases. Access to Web of Science will be provided by the University of Brighton (UK). Data Management support will be sought from COVIDENCE® systematic review software.
Sponsor (5b)/Funding	This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.
Role of sponsor/funder (5c)	Not Applicable.

Methodology for this Protocol
We followed the PRISMA-P guidelines which recommend items to address in a systematic review protocol.

Abstract

Introduction: Ischemic Heart Disease (IHD) is one of the leading causes of death & disease burden in India affecting all age groups. To reduce the deaths and tackle the burden of existing IHD the government approach has been mostly through the National Health Policy (2017) and National Program for Prevention & Control of Diabetes, Diabetes, Cardiovascular diseases and Stroke (NPCDCS). This paper offers a protocol for the systematic review of studies exploring the factors influencing service readiness of the public health system of India to tackle the burden of IHD.

Methods and Analysis: : Electronic databases of EMBASE (Ovid), AMED(Ovid), HMIC(Ovid), BNI(ProQuest), CINAHL(EBSCO), EMCARE(Ovid), PsycINFO(ProQuest), Medline/PubMed and Web of Science (Clarivate Analytics) will be searched till 1st October, 2020 for primary studies. Grey literature will be accessed through OpenGrey, TRIP Medical, WHO Database, MoHFW Website, OGD Platform India and Google Scholar (between 2010-2020). Primary studies meeting the eligibility criteria & grey literature published in English between 2010 to 2020 will be included. Data will be analysed through a

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

conceptual framework and primary outcome will constitute both quantitative and qualitative data. The quality of included studies will be assessed based on study design. Data will be managed on the COVIDENCE® platform. All authors will be involved in data extraction, quality appraisal, data synthesis and formulation of the final draft.

Ethics and dissemination: This study being a systematic review does not involve any clinical trial, primary data collection or empirical study involving humans or animals. Therefore, no ethical permissions were sought by reviewers.

Registration details: This systematic review protocol is registered in the International Prospective register of Systematic Reviews (PROSPERO) with number CRD 42020219490.

Link: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=219490

Strengths and limitations of this study

- The study can inform policymakers, academicians, NGOs and healthcare administrators with understanding of the existing gaps in the health system.
- This study can provide options to optimise existing health system resources of India to extend better quality of care for the patients.
- Our use of a deductive approach with a conceptual framework for data analysis may exclude some findings which might be a limitation.
- India being a vast country with a decentralised system of health governance might have varied levels of service readiness across states making it difficult to generalise study findings for all contexts.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

INTRODUCTION

6. Rationale

Ischemic Heart Disease (IHD) is the leading cause of deaths and disease burden in the adults of India. In 2019, IHD alone claimed approximately 1.5 million deaths in India with a mortality rate of 109.23 deaths per 100000 populations**.[1]** Between 2009 and 2019, deaths due to IHD increased significantly by 29.8% in India and in the year 2019 contributed to 7.79% of the total Disability Adjusted Life Years (DALYs). IHD affected all age groups including the working population of the country, leading to their premature deaths. High burden of disease & deaths due to IHD can have significant socio-economic implications for India. Specifically, the working population of India who are earning members within households may suffer premature deaths, leading to loss of individual lives and resulting in economic instability for their dependents. Moreover, healthcare expenditure is majorly met through out of pocket (OOP) expenses, presently at 63% of current health expenditure (CHE) of India, pushing households towards financial impoverishment.**[2]** In such a situation, tackling the disease burden and reducing deaths due to IHD seems necessary to avert loss of individual lives and prevent possible financial impoverishment of their households.

In order to reduce the deaths and tackle the burden of IHD in India the government approach has been mostly through context centric tactics of tackling the Non-Communicable Diseases (NCDs) through nationwide health sector policy & programs.**[3]** In agreement with the Global Action Plan for Prevention & Control of NCDs 2013-2020, endorsed by the World Health Assembly (WHA) in 2013, India has adopted 25 indicators and 9 voluntary goals to reduce 25% of premature deaths due to NCDs by 2025.**[4,5]** Premature mortality associated with IHD is indicated as unconditional probability of dying due to IHD between the age bands of 30-70 years. In India, the translation of this global action plan at national level has occurred through the National Health Policy (NHP-2017) and the policy vision is being implemented across states through multiple approaches by the National Program for Prevention & Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS).**[6,7]** Additionally, India's flagship program on universal health coverage (UHC), Ayushman Bharat, comprises the conversion of 150000 Primary Health Centres (PHCs) into Health & Wellness Centres (HWC). This measure aims to strengthen primary care services that can prevent & treat NCDs including IHD.**[8]** To support the cause, the NHP-2017 planned to raise the General Government Health Expenditure (GGHE) to 2.5% of the GDP from the usual government allocation of 1.2% of the GDP in previous years. Moreover, NHP-2017 has suggested NPCDCS to strengthen secondary & tertiary care services, along with primary care NCD interventions, such as 'Best Buys'.**[9]**

However, while the plans and commitments have been put in place, the challenges of the existing disease burden of IHD, upscaling premature deaths due to IHD, overwhelmed public healthcare system, scarcity of health workers and the poor co-operation between states and the central government in India's decentralized

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

health administrative structure appear to be detrimental towards tackling the disease burden of IHD.**[10-12]** Therefore, it becomes necessary to explore the factors influencing readiness of the public health system of India in order to tackle IHD based upon the commitments made through the NHP-2017 & other relevant national programs to tackle the increasing number of cases & deaths due to IHD. Readiness here is defined as the cumulative availability of components required to provide services.**[13]**

In a nutshell, this is an attempt to explore whether the existing public health system is capable of delivering the government commitments on national health policy and national NCD program to the people of India, through the existing public health system infrastructure, in order to tackle the mortality & morbidity due to IHD.

Our preliminary search on this topic in Cochrane Database of Systematic Reviews, PROSPERO, JBI Evidence Synthesis, CRD/DARE Database and Campbell Collaboration revealed no systematic reviews, scoping reviews, rapid reviews or published protocols related to India concerning readiness of health system. However, there were reviews focussed on health system components for Low and Middle Income Countries (LMIC) settings, acute care settings concerning LMIC countries, monitoring of cardiac patients and risk factors associated with IHD in Asian Indians (included those of Indian origin living outside India) which bear no similarity to our study.**[14-18]** Moreover, data from such studies considered LMIC context in a totality which made it difficult to find any specific data for India. The COVID-19 pandemic in India is already witnessing a scarcity of oxygen supply and overwhelmed tertiary care sector. **[19]** Further, health system issues like dual practice, poor skill mix and inadequate number of healthcare staff in the public sector has further constrained the existing care system.**[20]** In this scenario people with IHD or at risk with IHD might overstrain the existing care system especially with cases requiring tertiary care services.

7. Objectives

To identify the health system factors that contributes to facility readiness of the public healthcare system of India in tackling the burden of IHD. To achieve the stated objective a systematic review to analyse the existing peer reviewed studies and grey literature is proposed to be undertaken. The analysis will be guided through a conceptual framework, the 'WHO Framework for Action'. **[21]**

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

METHODS

8.Eligibility Criteria

i) Population, or participants and conditions of interest

Studies involving Indian people living with or at risk of IHD. Primary studies related to such population type published in peer reviewed journals and grey literature in English published between 2010 and 2020 will be included.

ii) Interventions or exposures

Relevant primary studies in peer reviewed journals & grey literature on national programs of NCD, National Health Policies (NHP-2012 & NHP-2017), interventions concerned with both prevention & control of Ischemic Heart Disease in India will be included if published in English between 2010 to 2020.

iii) Comparisons or control groups None.

iv) Outcomes of interest

Data of both quantitative & qualitative nature related to health system preparedness or readiness to prevent & tackle the burden of Ischemic Heart Disease in India.

v) Setting

Primary, Secondary & Tertiary Care settings in Public Health System of India.

vi) Study Designs

All forms of study design of both quantitative & qualitative nature will be included.

vii) Time Frame: 2010-2020. This timeframe has been chosen considering that the NPCDCS was launched by the Ministry of Health & Family Welfare, Government of India, in 2010. **[22]**

viii) Reporting Characteristics:

Years Considered: 2010 to 2020; Language: English; Publication Status: Peer Reviewed.

9. Information Sources: Details of Information sources has been outlined in ***Supplementary Material A: Information Sources & Representative Search Strategy.***

10. Search Strategy: Electronic databases of EMBASE (Ovid), AMED(Ovid), HMIC(Ovid), BNI(ProQuest), CINAHL(EBSCO), EMCARE(Ovid), PsycINFO(ProQuest), Medline/PubMed and Web of Science (Clarivate Analytics) will be searched with use of appropriate Boolean operators (AND, OR & NOT) till 2020. No filters or limits will be used at this stage. All electronic databases (except Medline/PubMed and Web of Science) will be accessed through the Healthcare Databases for NHS in England (HDAS) of the NHS Openathens, UK. HDAS mirrors search results of platforms like Ovid, EBSCO and ProQuest to retrieve results from these databases.

Trial registers like PROSPERO, Cochrane Database of Systematic Reviews, Campbell Collaboration and CRD/DARE Database will be searched to avoid duplication of existing knowledge on the topic and identify relevant primary studies from available systematic reviews.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Moreover, citation tracking, footnote chasing, bibliography scan and citation alerts (on web of science) will be conducted for a comprehensive search.**[23]** A combination of free texts and corresponding synonyms of terms comprising but not limited to "Ischemic Heart Disease" AND "Health System" AND "Readiness" AND "India" will be made for the initial search and appropriate preidentified MeSH Terms ('Emtree Terms' for Embase) in PubMed/Medline will be included to make a comprehensive search strategy across the electronic databases (Supplementary Table 1 added as a representative example). Thesaurus terms appropriate to the databases will be used to account for the variation in such terms across databases.

Grey literature will be accessed through OpenGrey, TRIP Medical, WHO Database, MoHFW Website, OGD Platform India and Google Scholar (between 2010-2020).

Detailed list of information sources & representative search result for this study are given in ***'Supplementary Material A: Information Sources & Representative Search Strategy'.***

STUDY RECORDS

11a. Data management

COVIDENCE® systematic review software (available at : www.covidence.org) will be employed to manage the studies retrieved and initiate the process of systematic review. On this software duplicate articles will be rejected in order to gather a set of unique articles for this study.**[24]** All article screening activities, selection of final set of articles for inclusion in the study after examination of full texts of eligible studies, data extraction and risk of bias assessment will be carried out on COVIDENCE®. References will be managed and stored on Endnote®. All forms of article selection, data extraction, quality appraisal, risk of bias assessment will be performed and a 'PRISMA flow chart', based on PRISMA guidelines for systematic review, will be created on the COVIDENCE® platform.

11b. Selection process

Initially, all duplicate articles will be checked by three reviewers (SB, NY, CLP) individually through a double blinded process to assure a rigorous review approach and rejected before identifying a unique set of articles for this study on the COVIDENCE® platform.

Afterwards, the screening of title and abstracts of potential articles will be conducted by the same reviewers in a similar double blinded manner to assure rigor of the study.**[25]** In case of any conflict related to the inclusion of an article, it will be temporarily included in the next step for further evaluation.

In the next step, full-text articles will be examined based on the inclusion and exclusion criteria before final inclusion in the study. Reasons for the inclusion and exclusion of articles will be clearly identified and documented by each reviewer independently on ***Supplementary Material B*** (Form for Screening of full text articles based on inclusion & exclusion criteria with documented justification). Assuring three reviewers will allow for a consensus-based conflict resolution in case there is uncertainty raised about inclusion of an

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

article for the study. Moreover, the fourth researcher will also be consulted before finalising the final set of included articles for this study.

11c. Data Collection Process

Data will be extracted based on ***Supplementary Material C*** (Data Extraction Form for Included Studies) by three reviewers (SB, NY & CLP) independently in a blind manner for each included study. SB will randomly select half of the included studies which will be given to the fourth researcher to extract data independently. This will ensure rigor of the study and a consensus-based approach for approving the final content of the data extraction form. Any disagreements will be resolved through mediation with the fourth researcher. The data extraction form will also offer content on the risk of bias and quality assessment of included studies. As the final data synthesis will involve using a conceptual framework, all relevant quantitative data will be converted into meaningful qualitative data for analysis and interpretation.

12. Data Items

The descriptive data items that will be collected constitute : (i) Characteristics of the study including country & year of publication, research aim & participant characteristics; (ii) Methodological characteristics including study design, data/information collection methods and data analysis; (iii) Results associated with readiness of the public health system to tackle the burden of ischemic heart disease along with concluding remarks from the article. The data collection items will also include risk of bias and quality assessment of included individual articles based on appropriate tools. The proposed tools and process for assessment of risk of bias & quality assessment of eligible studies is further detailed in the section pertaining to 'risk of bias in individual studies' within this protocol. For this study we wish to create simplification of quantitative data by transforming them to meaningful qualitative units to accommodate them into the conceptual framework opted for our study. Moreover, only relevant data will be extracted based on the needs of the study.

We will extract data from those studies concerned with primary care facilities where facility assessment can provide data pertaining to facility readiness even if readiness concerning ischemic heart disease is not mentioned. This is because treatment of Ischemic Heart Disease and its risk factors are integrated at the primary care level along with treatment of other communicable and non-communicable diseases.**[26]**

13. Outcomes and prioritization

Outcome of this study will constitute both quantitative and qualitative data retrieved from the included articles to inform health system factors that influence the readiness of India's public health system to tackle the burden of Ischemic Heart Disease.

This will include data at all levels of care that informs health service delivery, health workforce, health information system, access to essential medicines, health financing and health system governance concerning the present topic.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

In terms of quantitative data, data from facility readiness assessment, health facility assessment, WHO Service Availability & Readiness Assessment (SARA) data, surveys, baseline study of health facilities and monitoring & evaluation data of health facilities along with data from the national NCD program of India will serve to provide information on the outcome.

In terms of qualitative studies, data from in-depth interviews, focus group discussions, researcher observations of facilities and data from the national health policies will serve to understand the outcomes better.

14. Risk of bias in individual studies

All eligible studies at the outcome stage are expected to have methodological variation in approach and therefore, we will assess their quality & risk of bias at this stage. In order to minimize the risk of bias and evaluate the quality of included studies the Cochrane collaboration tool for assessing the risk of bias of Randomized Control Trials (RCTs), Newcastle-Ottawa Scale for cohort & case-control studies, NIH Quality Assessment Tool for Cross-Sectional Studies and Critical Appraisal Skills Program (CASP) tool for qualitative studies will be used based on the methodological design of the individual eligible study. **[27-30]** We will not reject any study based on assessed quality only the information will be documented in the data extraction and synthesis stage. The risk of bias assessment and quality appraisal will be independently performed by each reviewer before converging to a mutually agreed decision. Any discrepancies will be mediated through the fourth researcher.

National Health Policies of India, national NCD program guidelines and raw primary data from the OGD Platform cannot be assessed for risk of bias and quality appraisal. They will be excluded from this process of evaluation.

15. DATA SYNTHESIS

15 (a &b) Data Synthesis

The extracted data will be analysed through a conceptual framework, the 'WHO Framework for Action', followed by summarization of findings to answer the research questions. We do not expect to conduct a meta-analysis as we do not anticipate many articles in which meta-analysis may be appropriate. Additionally, the inclusion of qualitative studies will likely lead to a heterogeneous outcome which is not suitable for meta-analysis. Moreover, we will be transforming the quantitative data to meaningful qualitative units for assimilation into the conceptual framework for our study. Data analysis through the proposed conceptual framework rather serves as a conceptual lens for this study. Therefore, we will be deductive in our approach assuming all health system readiness issues are inclusive within this conceptual framework. The conceptual framework was chosen through mediation with all reviewers and moreover, straight forward demarcation of health system components allows for clear interpretation.**[31]** We also considered other

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

conceptual frameworks; however, they were either overly complex for the purposes of this study or considered factors beyond the health system that were not relevant for the present study.**[32-36]**

15 (c) Sub-Group Analysis

We anticipate that sub-group analysis might be performed from the extracted data by using the components of 'Health System Strengthening' (HSS) building blocks from the proposed conceptual framework.

15 (d) Data Summary

Data will be summarized thematically based on 'Health System Strengthening' (HSS) building blocks from the proposed conceptual framework.

16. Meta-Bias(es)

We do not plan to conduct any assessment of meta-bias(es) as the expected studies included for this systematic review might be of varied study design making it difficult to assess such bias such as 'publication bias across studies' or 'selective reporting within studies'. **[37-38]**

In this context, our approach to this review might be influenced by 'language bias' as we are including studies published only in English. We plan to mention this aspect as our study limitation in the final review. Further, performing citation tracking can lead us to a "biased sample of studies". **[39]** We plan to mitigate this by involving multiple authors in the screening, data extraction and data analysis process. Moreover, using a particular conceptual framework with a deductive research approach might lead us to a selective reporting of outcomes. We plan to avoid this through collective discussions between all the researchers post data extraction stage.

17. Confidence in cumulative evidence

At present we do not have plans to assess the confidence in cumulative evidence as this study does not report effectiveness of clinical outcome(s).

18. Presenting or reporting the results

The proposed systematic review will transparently represent the entire screening process and offer information on the finally included studies through a flow diagram in accordance with the PRISMA-P guidelines**.[40]** Data collection will be through a data extraction form with outcomes to extract data related to health system readiness. Data will be summarized in tables with details of study characteristics, methodological characteristics and individual study results. We will also provide data of included and excluded studies with documented justification of their inclusion or exclusion.

19. Patient and public involvement

The present study is a systematic review not an empirical study and therefore, there is no direct involvement of patient or public in this study.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

20. Ethics & Dissemination

20 (a) Ethical Issues

This study being a systematic review does not involve any clinical trial, primary data collection or empirical study involving humans or animals. Therefore, no ethical permissions were sought by reviewers.

20 (b) Publication Plan

We plan to publish the preliminary results of this review in conferences and proceed on to publish the final article in peer reviewed journal once completed. We hope to publish both the protocol & systematic review in open access journals to benefit academic communities, policymakers and other public health stakeholders with the ambition to improve the readiness of the health to tackle the burden of Ischemic Heart Disease in India and elsewhere.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

References:

1. Institute for Health Metrics and Evaluation (IHME). Findings from the Global Burden of Disease Study 2019. Seattle, WA: IHME, 2021. Available from: https://vizhub.healthdata.org/gbd-compare/
2. World Health Organization (WHO). Global health expenditure database. Last updated August 20, 2020. http://www.who.int/health-accounts/ghed/en/ (accessed Sept. 18, 2020).
3. Ministry of Health & Family Welfare (MoHFW, India). National Health Portal (2019, November 5). Non-communicable Diseases. Retrieved from https://www.nhp.gov.in/healthlyliving/ncd2019 (accessed Sept 18, 2020).
4. WHO. Global action plan for the prevention and control of noncommunicable diseases 2013-2020. World Health Organization. 2013 Mar 29:102.
5. Ministry of Health & Family Welfare (MoHFW, India). National Action Plan and Monitoring Framework for prevention & control of Noncommunicable Diseases (NCDs). Retrieved from https://www.iccp-portal.org/system/files/plans/India%20-%20National_Action_Plan_and_Monitoring_Framework_Prevention_NCD_2013.pdf (accessed on Sept 18, 2020).
6. Ministry of Health & Family Welfare (MoHFW, India). National Health Policy; 2017. Available from: http://www.cdsco.nic.in/writereaddata/National-Health-Policy.pdf (accessed on Sept. 18, 2020).
7. Ministry of Health & Family Welfare (MoHFW, India). Directorate General of Health Services (DGHS). National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke. Retrieved September 18, 2020, from https://dghs.gov.in/content/1363_3_NationalProgrammePreventionControl.aspx
8. Gopichandran V. Ayushman Bharat National Health Protection Scheme: An Ethical Analysis. Asian Bioethics Review 2019;11:69-80. doi:10.1007/s41649-019-00083-5
9. World Health Organization. Tackling NCDs: 'best buys' and other recommended interventions for the prevention and control of noncommunicable diseases. World Health Organization; 2017.
10. Narain JP. Public health challenges in India: seizing the opportunities. Indian journal of community medicine: official publication of Indian Association of Preventive & Social Medicine. 2016 Apr;41(2):85.
11. Tiwari R, Negandhi H, Zodpey S. Health Management Workforce for India in 2030. Frontiers in Public Health 2018;6. doi:10.3389/fpubh.2018.00227

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12. Lakshminarayanan S. Role of government in public health: Current scenario in India and future scope. Journal of Family and Community Medicine 2011;18:26. doi:10.4103/1319-1683.78635
13. Abazinab S, Woldie M, Alaro T. Readiness of health centers and primary hospitals for the implementation of proposed health insurance schemes in Southwest Ethiopia. Ethiopian Journal of Health Sciences 2016;26:449. doi:10.4314/ejhs.v26i5.6
14. Marin T, Walsh S, May N et al. Screening for depression and anxiety among patients with acute coronary syndrome in acute care settings: a scoping review. JBI Evidence Synthesis 2020;18:1932-1969. doi:10.11124/jbisrir-d-19-00316
15. Likka M, Handalo D, Weldsilase Y et al. The effect of community-based health insurance schemes on utilization of healthcare services in low- and middle-income countries: a systematic review protocol of quantitative evidence. JBI Database of Systematic Reviews and Implementation Reports 2018;16:653-661. doi:10.11124/jbisrir-2017-003381
16. Teferra M, Ramos J, Kourbelis C et al. Electronic textile-based electrocardiogram monitoring in cardiac patients. JBI Database of Systematic Reviews and Implementation Reports 2019;17:1958-1998. doi:10.11124/jbisrir-2017-003989
17. Ninan A, Acosta J, Kulesza T et al. A systematic review of the effectiveness of preventive measures for coronary artery disease in Asian Indians. JBI Library of Systematic Reviews 2011;9:1-17. doi:10.11124/jbisrir-2011-314
18. Ninan A, Acosta J, Kulesza T et al. A systematic review on the risk factors of coronary artery disease found in Asian Indians. JBI Library of Systematic Reviews 2011;9:1-14. doi:10.11124/jbisrir-2011-455
19. Kumar P, Kumar M. Management of potential ventilator shortage in India in view of on-going COVID-19 pandemic. Indian Journal of Anaesthesia 2020;64:151. doi:10.4103/ija.ija_342_20
20. Iyengar KP, Jain VK, Vaishya R. Current situation with doctors and healthcare workers during COVID-19 pandemic in India. Postgraduate Medical Journal Published Online First: 19 August 2020. doi: 10.1136/postgradmedj-2020-138496
21. World Health Organization. Everybody's business--strengthening health systems to improve health outcomes: WHO's framework for action. 2007.
22. Arokiasamy P. India's escalating burden of non-communicable diseases. The lancet global health. 2018 Dec 1;6(12):e1262-3.
23. Wright K, Golder S, Rodriguez-Lopez R. Citation searching: a systematic review case study of multiple risk behaviour interventions. BMC Medical Research Methodology 2014;14. doi:10.1186/1471-2288-14-73
24. Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia. Available at www.covidence.org .2021.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

25. Ayala Quintanilla B, Taft A, McDonald S et al. Social determinants and maternal exposure to intimate partner violence of obstetric patients with severe maternal morbidity in the intensive care unit: a systematic review protocol. BMJ Open 2016;6:e013270. doi:10.1136/bmjopen-2016-013270
26. Ministry of Health & Family Welfare, Government of India. National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Disease and Stroke: Operational Guidelines. Strategy. 2010.
27. Higgins J, Altman D, Gotzsche P et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. BMJ 2011;343:d5928-d5928. doi:10.1136/bmj.d5928
28. Peterson J, Welch V, Losos M, Tugwell PJ. The Newcastle-Ottawa scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Ottawa: Ottawa Hospital Research Institute. 2011.
29. National Institutes of Health (NIH). Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies. 2014. Available online at: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools (Accessed November 19, 2020).
30. Critical Appraisal Skills Programme. (CASP) qualitative checklist, 2018, https://casp-uk.net/wp-content/uploads/2018/01/CASP-Qualitative-Checklist-2018.pdf (accessed 18 Sept. 2020)
31. Mounier-Jack S, Griffiths U, Closser S et al. Measuring the health systems impact of disease control programmes: a critical reflection on the WHO building blocks framework. BMC Public Health 2014;14. doi:10.1186/1471-2458-14-278
32. Roberts, M.J., W.C. Hsiao, P. Berman, and M.R Reich. 2003. Getting Health Reform Right. New York: Oxford University Press, page 27
33. De Savigny, D. and Adam, T. (eds.) 2009, Systems thinking for health systems strengthening, World Health Organization, Geneva.
34. Shakarishvili G, Lansang M, Mitta V et al. Health systems strengthening: a common classification and framework for investment analysis. Health Policy and Planning 2010;26:316-326. doi:10.1093/heapol/czq053
35. Josefien van Olmen, Bart Criel, Wim Van Damme, Bruno Marchal, Sara Van Belle, Monique Van Dormael, Tom Hoerée, Marjan Pirard and Guy Kegels (2010). Analysing health systems to make them stronger.ITG Press, Antwerpen, Belgium.
36. Murray, C. J., & Frenk, J. (2000). A framework for assessing the performance of health systems. Bulletin of the world Health Organization, 78, 717-731.
37. Dalton J, Bolen S, Mascha E. Publication Bias. Anaesthesia & Analgesia 2016;123:812-813. doi:10.1213/ane.0000000000001596
38. Reid E, Tejani A, Huan L et al. Managing the incidence of selective reporting bias: a survey of Cochrane review groups. Systematic Reviews 2015;4. doi:10.1186/s13643-015-0070-y

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

39. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions version 6.2 (updated February 2021). Cochrane, 2021. Available from www.training.cochrane.org/handbook.
40. Moher D, Liberati A, Tetzlaff J et al. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Medicine 2009;6:e1000097. doi:10.1371/journal.pmed.1000097

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Material A: Information Sources & Representative Search Strategy			
(i) Electronic Databases:	**Database**	**Accessed From**	**Planned Dates of Coverage**
	Pubmed	NIH Website	From 2010 till 01.10.2020
	Embase	NHS HDAS Platform	From 2010 till 01.10.2020
	EMCARE	NHS HDAS Platform	From 2010 till 01.10.2020
	HMIC	NHS HDAS Platform	From 2010 till 01.10.2020
	BNI	NHS HDAS Platform	From 2010 till 01.10.2020
	CINAHL	NHS HDAS Platform	From 2010 till 01.10.2020
	AMED	NHS HDAS Platform	From 2010 till 01.10.2020
	PsycINFO	NHS HDAS Platform	From 2010 till 01.10.2020
	Web of Science	Clarivate Analytics	From 2010 till 01.10.2020
(ii)Trial Registers	PROSPERO	CRD Database	From 2010 till 01.10.2020
	Cochrane Database of Systematic Reviews	Cochrane Library	From 2010 till 01.10.2020
	Campbell Collaboration	Cambell Collaboration Website.	From 2010 till 01.10.2020

For peer review only

	JBI Evidence Synthesis	Joanna Briggs Institute Website.	From 2010 till 01.10.2020
	CRD/DARE Database	Centre for Reviews and Dissemination Website.	From 2010 till 01.10.2020.
(iii) Contact with study authors	Researchgate	Researchgate Website	As per necessity to request for full text articles based on articles published till 01.10.2020.
(iv) Grey Literature Sources	OpenGrey	INIST-CNRS managed Website	01.01.2010 to 01.10.2020
	TRIP Medical	TRIP Database Website	01.01.2010 to 01.10.2020
	WHO Database	World Health Organization Website	From 2010 till 01.10.2020.
	MoHFW	Ministry of Health & Family Welfare Website.	From 2010 till 01.10.2020.
	OGD Platform India	Government of India Website on Open Government Data.	From 2010 till 01.10.2020.
	Google Scholar	Google Scholar Website	From 2010 till 01.10.2020.

For peer review only

Table 1		
A representative search for CINAHL database performed on NHS Openathens HDAS Platform on 14th May 2021. Time frame: 2010-2020.Peer Reviewed. English Language.		
Sl. No.	**Search Terms**	**Results**
1	(India*).ti,ab OR (South Asia*).ti,ab OR (Asia*).ti,ab OR (LMIC).ti,ab OR (LIC).ti,ab OR (Low Income Countr*).ti,ab OR (Low Middle Income Countr*).ti,ab OR (SAARC).ti,ab OR (ASEAN).ti,ab	85642
2	(NCD*).ti,ab OR (non-communicable diseas*).ti,ab OR (chronic disease* OR myocardial ischemia).ti,ab OR (cardio?vascular diseas*).ti,ab OR (ischemic heart diseas* OR heart disease*).ti,ab OR (vascular disease*).ti,ab OR (Diabetes).ti,ab OR (Hypertension).ti,ab OR ("High Blood Pressure").ti,ab OR (Cancer).ti,ab OR (COPD).ti,ab OR (Chronic Obstructive Pulmonary Disease).ti,ab OR (Coronary Artery Diseas*).ti,ab OR (Coronary Heart Diseas*).ti,ab	789662
3	(health system* OR program*).ti,ab OR (health?care system*).ti,ab OR (health planning* OR health service utiliz*).ti,ab OR (health?care planning*).ti,ab OR (health facilit*).ti,ab OR (National Health Policy).ti,ab OR (National Health Program*).ti,ab OR (NPCDCS).ti,ab OR "primary care" OR "primary healthcare" OR "secondary care" OR healthcare OR "tertiary care" OR hospital*	1503977
4	(prepare* OR health system prepared*).ti,ab OR (ready).ti,ab OR (readiness).ti,ab OR (facility? assess*).ti,ab OR (readiness).ti,ab	77767
5	exp INDIA/ OR exp "ASIA, WESTERN"/	128168
6	exp ASIANS/	32187
7	exp "DEVELOPING COUNTRIES"/	18662
8	exp "LOW- AND MIDDLE INCOME COUNTRIES"/	1361
9	(5 OR 6 OR 7 OR 8)	176910

For peer review only

10	(1 OR 9)	225727
11	exp "CARDIOVASCULAR DISEASES"/ OR exp "HEART DISEASES"/ OR exp "VASCULAR DISEASES"/	584557
12	exp "CORONARY DISEASE"/	54503
13	exp "MYOCARDIAL ISCHEMIA"/ OR exp "ACUTE CORONARY SYNDROME"/	101965
14	exp "CHRONIC DISEASE"/ OR exp "NONCOMMUNICABLE DISEASES"/	64096
15	exp "DIABETES MELLITUS"/	164718
16	exp HYPERTENSION/	78476
17	exp "LUNG DISEASES, OBSTRUCTIVE"/	60675
18	exp NEOPLASMS/	544583
19	(11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18)	1311975
20	(2 OR 19)	1554463
21	(10 AND 20)	50731
22	exp "HEALTH AND WELFARE PLANNING"/ OR exp "HEALTH FACILITY PLANNING"/	171562
23	exp "HEALTH CARE DELIVERY"/	338701
24	exp "NATIONAL HEALTH PROGRAMS"/ OR exp "HEALTH RESOURCE UTILIZATION"/ OR exp "HEALTH SERVICES ACCESSIBILITY"/ OR exp "MANAGED CARE PROGRAMS"/ OR exp "PRIMARY HEALTH CARE"/ OR exp "SECONDARY HEALTH CARE"/ OR exp "TERTIARY HEALTH CARE"/	237596
25	exp "PUBLIC HEALTH ADMINISTRATION"/ OR exp "PROGRAM DEVELOPMENT"/	88044
26	exp "PUBLIC POLICY"/ OR exp "HEALTH POLICY"/	106774
27	exp "HEALTH POLICY STUDIES"/ OR exp "POLICY STUDIES"/	827
28	(22 OR 23 OR 24 OR 25 OR 26 OR 27)	526561
29	(3 OR 28)	1709631
30	(21 AND 29)	20354

For peer review only

31	(4 AND 30)	423
32	(4 AND 30) [DT 2010-2020] [Peer reviewed] [Languages eng]	326

For peer review only

Supplementary Material B: Form for Screening of full text articles based on inclusion & exclusion criteria with documented

Basic Inclusion Criteria								SB		CLP		NY		
Sl. No.	Title	Is this peer reviewed?	Is this paper in English?	Is this a primary study?	Published between 2000 to 2020?	Does it concern India?	Article related to the topic?	Reasons for Inclusion	Reasons for Exclusion	Reasons for Inclusion	Reasons for Exclusion	Reasons for Inclusion	Reasons for Exclusion	Final Decision (Yes/No)
1														
2														
3														
4														
5														
6														
7														
8														
9														
10														

For peer review only

Supplementary Material C: Data Extraction Form for Included Studies									
Sl. No.	Title	Author	Year & Country	Research Aims	Study Design	Participants	Methods	Results	Conclusions
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									

For peer review only

Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

For peer review only

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	1

Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	1
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1-2
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	2
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	3
Support			
Sources	#5a	Indicate sources of financial or other support for the review	3
Sponsor	#5b	Provide name for the review funder and / or sponsor	3
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	3
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is	5-6

For peer review only

		already known	
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7-8
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Study records - data collection	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any	9

For peer review only

process		processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	9
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	9-10
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	10
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	10-11
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	N/A
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	11
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	11
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within	11

For peer review only

		studies)	
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	11

The PRISMA-P checklist is distributed under the terms of the Creative Commons Attribution License CC-BY 4.0. This checklist was completed on 29. November 2020 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

For peer review only